



University of Antwerp
Faculty of Applied Engineering

CPS Project

Water-Quality Monitoring and Pollution Source Detection Buoy

Final Presentation

Why does this matter?

7 Million

people in Belgium and the Netherlands rely on the Meuse for their drinking water

67

substances exceeded EU target values in the Meuse in 2024

6,169 h

of disrupted drinking-water intake along the Meuse in 2024 due to high pollution

Faster mitigation starts with source localization.



A Quick Recap



Our Mission

Detect and localize pollution sources in the Meuse River using a **semi-autonomous drifting buoy + digital twin.**

Success criteria

4 h

Time to detect a contamination source from a single deployment

45 s

Maximum latency from pollution detection to alert

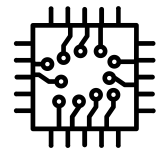
100 m

minimum source-localization accuracy along the river

<€1k

Hardware cost per buoy unit

What we promised



Physical

Sensing buoy

- pH sensor
- EC sensor
- DO
- GPS



Telemetry

Cloud link

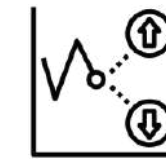
- 4G/LTE
- IoT platform



Cyber

Digital twin

- River model
- Pollution Source prediction



Decision

Source map

- State estimation
- Backtracking

Requirements formalized in STL

Functional Requirements (10)

FR1. Real-time monitoring

$G(\text{in_river} \rightarrow F[0,10s](\text{stable}(\text{pH}) \wedge \text{stable}(\text{EC}) \wedge \text{stable}(\text{DO})))$

FR5. GPS localization

$G(\text{data_sample} \rightarrow \text{gps_error} < 5\text{m} \wedge \text{timestamp_available})$

FR8. Automatic anomaly alerts

$G(\text{threshold_exceeded} \rightarrow F[0,5s](\text{alert_sent}))$

Non-Functional Requirements (6)

NFR1 — Failure detection

$G(\text{failure_detected} \rightarrow F[0,30s] \text{ user_notified})$

NFR3 — Power autonomy

$G[0,4h] (\text{battery_level} > 0)$

NFR6 — Cost constraint

$\text{TotalCost} \leq \text{€}1000$

Major decisions we made



Sensors

Compared: Hobbyist vs Industrial grade

Decision: DFRobot Gravity series

Why: 3x accuracy → satisfy FR8 Requirement



Platform

Compared: Stationary vs. Fully autonomous vs. Semi-autonomous

Decision: Semi-autonomous drifting

Why: Power-efficient, matches river physics, enables source tracing



Motors

Compared: Brushed DC vs. DC Gear vs. BLDC

Decision: BLDC

Why: More efficient, less heat, and better speed and torque control.



Communication

Compared: Cellular vs. Wi-Fi vs. LoRa vs. Bluetooth

Decision: 4G/LTE with GNSS

Why: No base station needed, sufficient bandwidth, area covered

Our 14-week journey

Week 1-4

- Make a work plan
- Divide the tasks
- Order the sensors
- Wait for the order shipment

Week 5-6

- pH/EC calibrated
- DO issue discovered
- LTE/GNSS tested
- Thingsboard dashboard
- Digital Twin

Week 7-8

- Meuse-specific requirements
- Buoy-shape trade-off
- RC bait boat selected.

Week 11

- PCB integration failed
- Abandoned reverse engineering
- Found a new solution

Week 12-13

- System rebuilt around ESP32
- Custom PCB
- 3D-printed mount
- SD logging
- IMU sensor addition
- digital twin integration

Week 14

- Pond test with real GNSS,
- Thingsboard alarms,
- thresholds, and sensor readings.

Sensors



Sensors

EC sensor



DO Sensor



pH Sensor



1. Wrote Firmware against datasheets
2. Tested with buffer solutions



Total Cost: € 310



Sensors

DO sensor “surprise”

1. No membrane cap fluid
2. No testing solution



Drie Eiken Campus

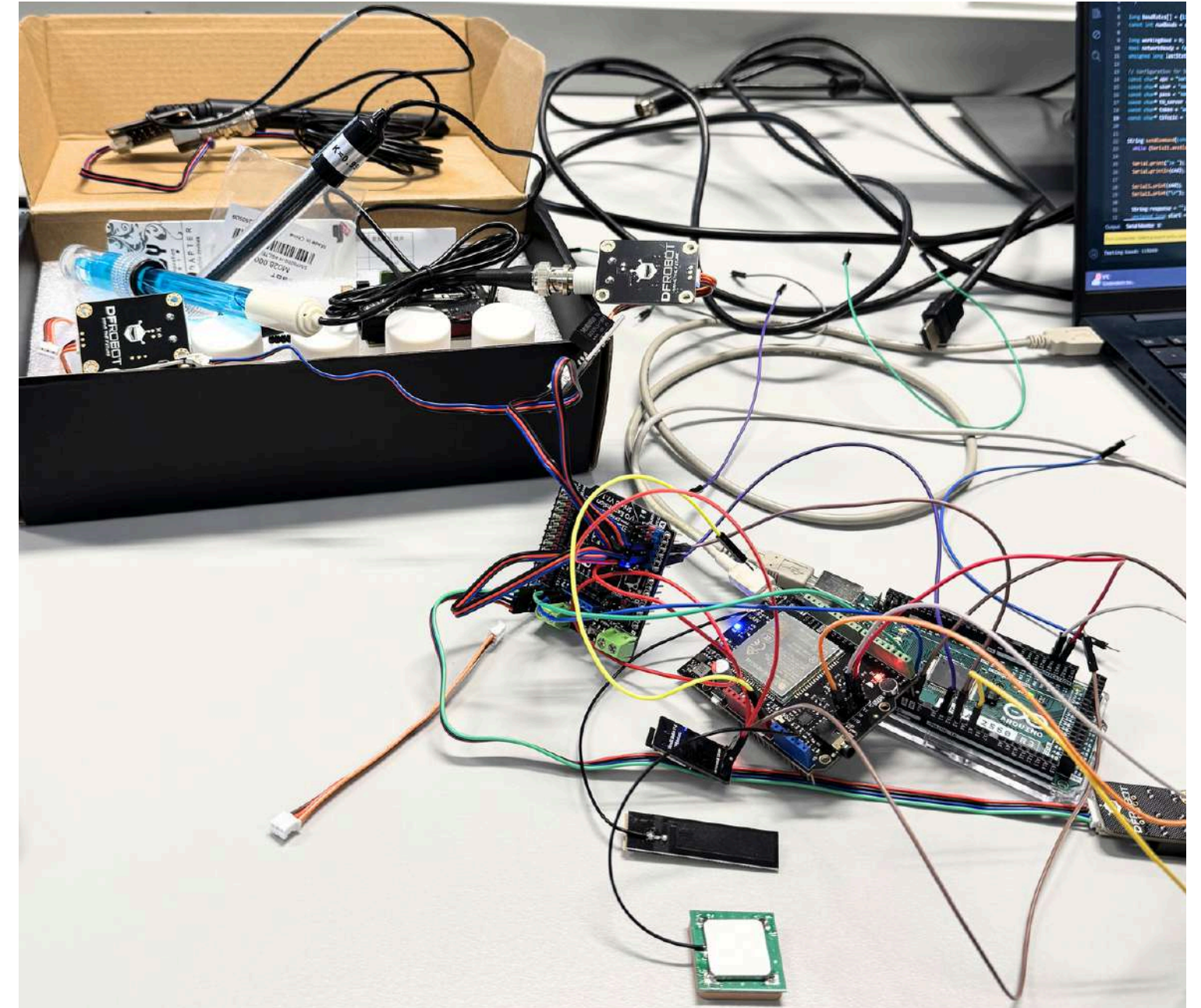


Testing

The testing we did



The mess



1

Soak the probes

30 minutes in pH 7 buffer to wet the glass membrane and stabilize

2

Two-point testing

Two buffer solutions per sensor

3

Verify with Requirements

Sensor reading accuracy
pH ± 0.1 , EC $\pm 8\%$, DO ± 0.2 mg/L confirmed

The Integration with Thingsboard

CPS

Time series table

History - last 70 days

Timestamp	DO ↓	EC	pH	latitude	longitude
2026-05-26 16:15:55	4.86	15.16	4.51		
2026-05-26 16:16:10	4.86	15.15	4.51		
2026-05-26 16:16:20	4.86	15.16	4.52		
2026-05-26 16:17:11	4.86	15.16	4.5		
2026-05-26 16:17:26	4.86	15.16	4.52		
2026-05-26 16:17:49	4.86	15.15	4.49		

Time series chart

- DO
- EC
- pH



Why Thingsboard?

We keep control

- open-source and self-hostable
- Not locked into a closed cloud service.

Built-in Alerts

- define alarm rules
- redundancy to the Digital Twin operation

Live Dashboard

- easier to test and visualize
- track communication performance

Multi-Device Support

- extendable to a fleet of buoys
- keep the system scalable



1 million Data points per month



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Ingenieurswetenschappen

Meuse River Thresholds

From generic thresholds to Meuse River Specifics

Consulted Research and Official reports

Parameter	Critical (low)	Warning (low)	Healthy/Meuse normal	Warning (high)	Critical (high)
pH	< 6.0	6.0 – 6.5	6.5 – 8.5 (Meuse: 7.5 – 8.2)	8.5 – 9.0	> 9.0
EC (µS/cm)		< 200 (atypical)	200 – 700 (limestone normal)	700 – 1000 (low-flow, runoff)	> 1000 polluted > 2000 severe
DO (mg/L)	< 3 anoxic	3 – 5 hypoxic	7 – 9 good 9 – 12 excellent	5 – 7 moderate	> 12 supersat.

Sources:
[1] [Van Vliet & Zwolsman, 2008](#);
[2] [Catteau et al., 2021](#);
[3] [RIWA-Maas, 2021](#);
[4] [EU WFD Directive, 2000](#).

FR8 Requirement Satisfied

Alarm Rules in Thingsboard

Alarm rule filter

☐	Created time ↓	Alarm type	Entity type	Entity	Severities
☐	2026-03-29 22:45:43	DO < 5.0	Device	CPSBuoy	Critical
☐	2026-03-29 22:43:52	5.0 < DO < 7.0	Device	CPSBuoy	Warning
☐	2026-03-29 20:16:41	EC > 1000 uS/cm	Device	CPSBuoy	Critical
☐	2026-03-29 20:15:07	700 < EC < 1000 uS/cm	Device	CPSBuoy	Warning
☐	2026-03-29 20:04:59	ph < 6.0	Device	CPSBuoy	Critical
☐	2026-03-29 20:03:16	6.5 < pH < 6.0	Device	CPSBuoy	Warning
☐	2026-03-29 20:01:07	pH > 9.0	Device	CPSBuoy	Critical

ent subscription

Free

Status

Active

3

sam.roels@student.uantwerpen.be

Tenant administrator

Notification

Mark all as read

configure

Alarm 'pH > 9.0' - acknowledged

Severity: critical, originator: Device 'CPSBuoy'

1 hr

Alarm 'pH > 9.0' - cleared

Severity: critical, originator: Device 'CPSBuoy'

1 hr

New alarm 'DO < 5.0'

Severity: critical, originator: Device 'CPSBuoy'

2 hr

Critical

View all



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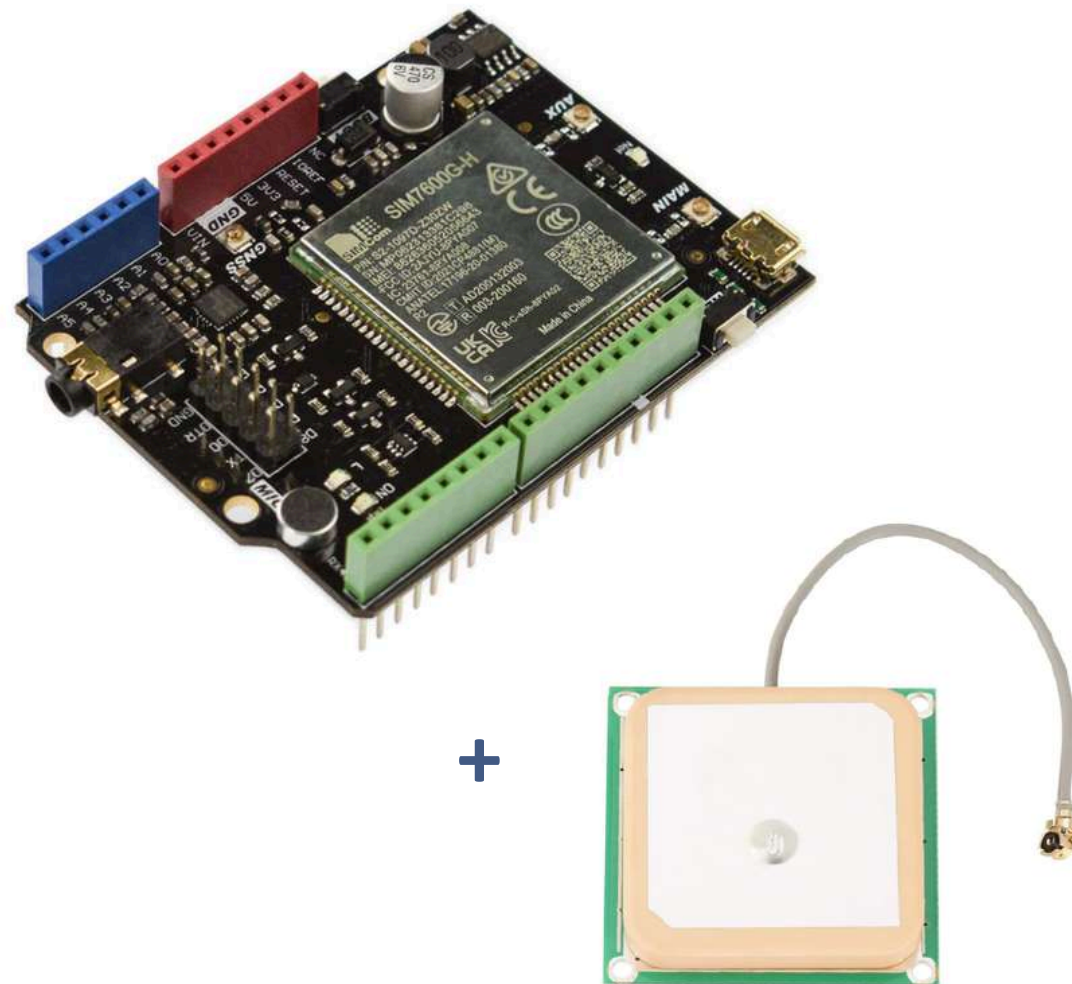
Communication



Communication

We tested and compared two communication shields:

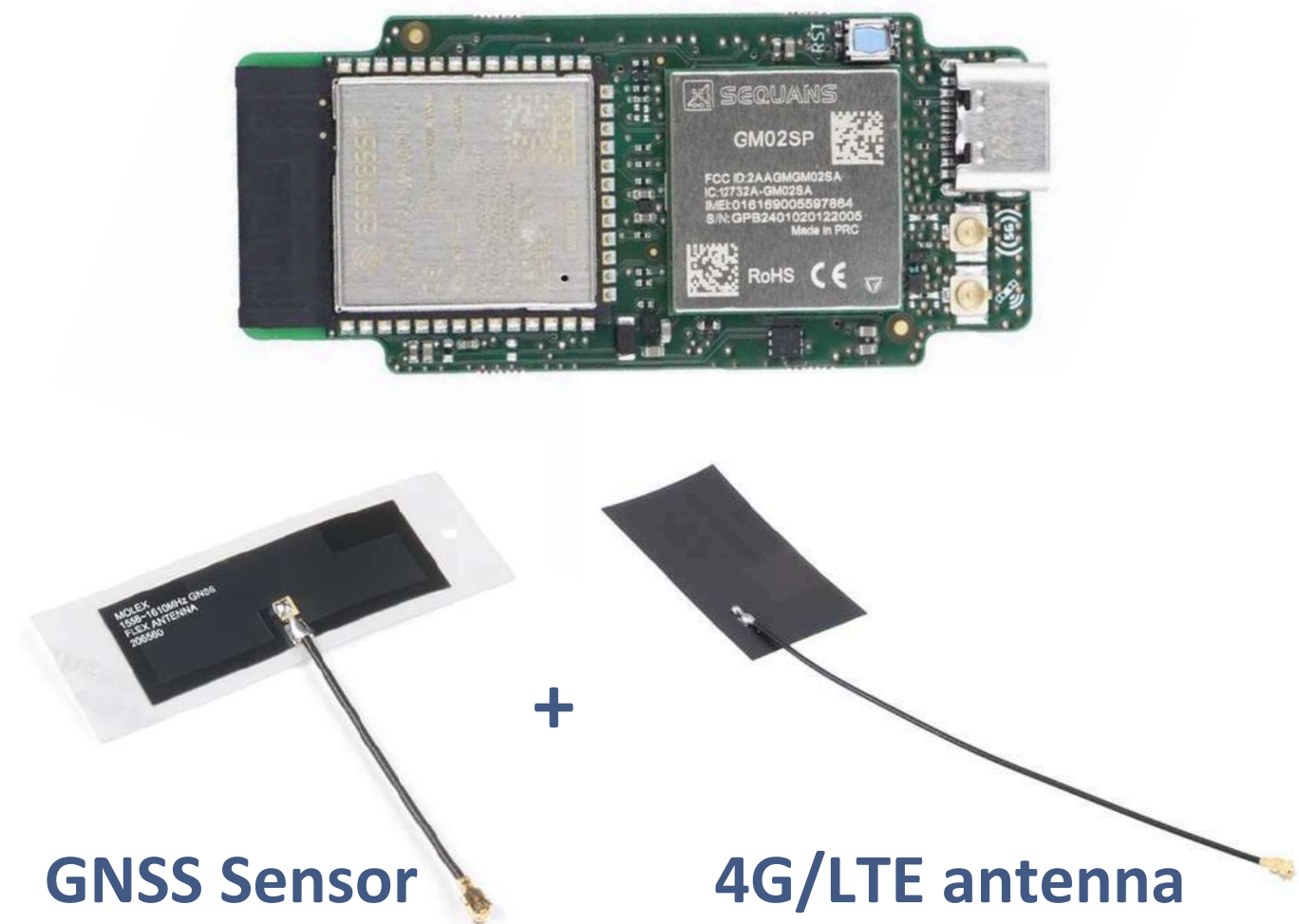
4G (LTE) Shield for Arduino



GPS Sensor

vs.

SoM with an ESP32-S3



4G/LTE Shield comparison

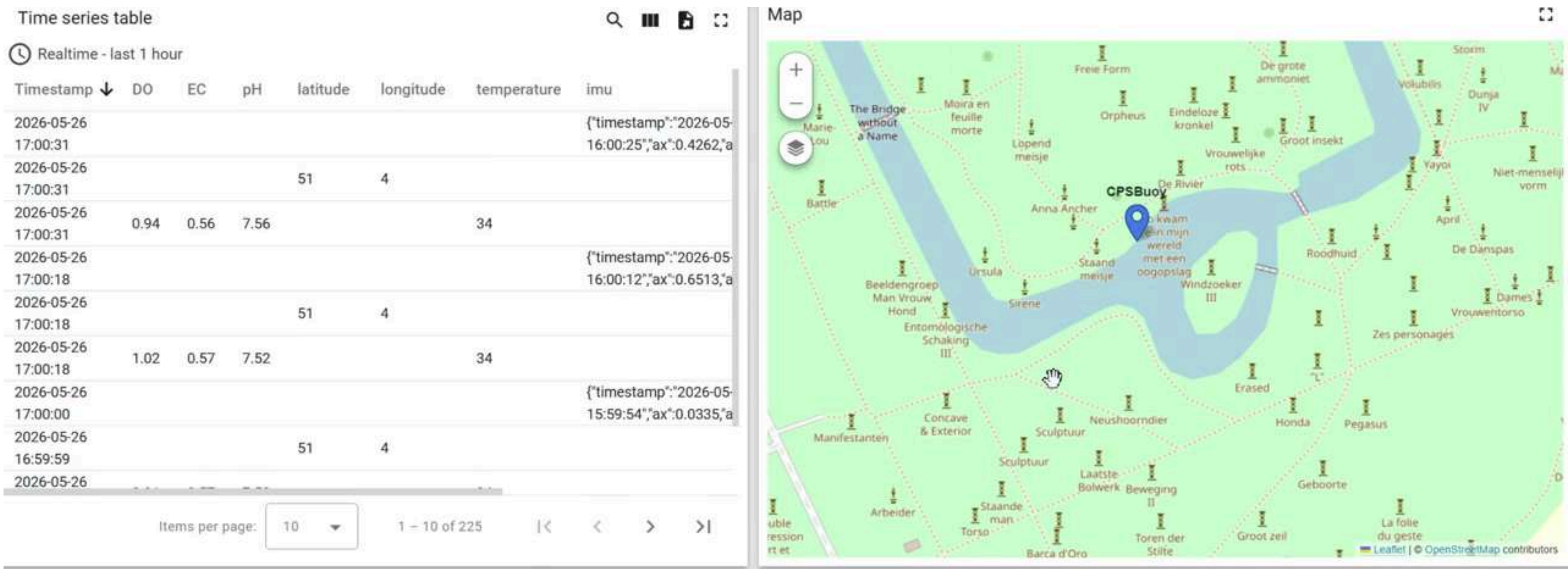
4G (LTE) Shield for Arduino		SoM with an ESP32-S3
GNSS fix time	> 5 min, often no fix	< 30 s cold start
4G handshake	Intermittent	First try, stable
Antenna setup	Single connector, cross-talk	Dual (GNSS + LTE) clean
MCU Included	Requires Arduino Support	Included ESP32-S3
Active Current	300 - 500 mA peak (LTE burst)	100 mA peak (LTE-M)

4G/LTE Shield comparison

4G (LTE) Shield for Arduino		SoM with an ESP32-S3
GNSS fix time	> 5 min, often no fix	< 30 s cold start
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Active Current	300 - 500 mA peak (LTE burst)	100 mA peak (LTE-M)

To satisfy FR5 (gps_error < 5 m) and keep the operation reliable we picked System-on-Modules Shield

Live water-quality dashboard

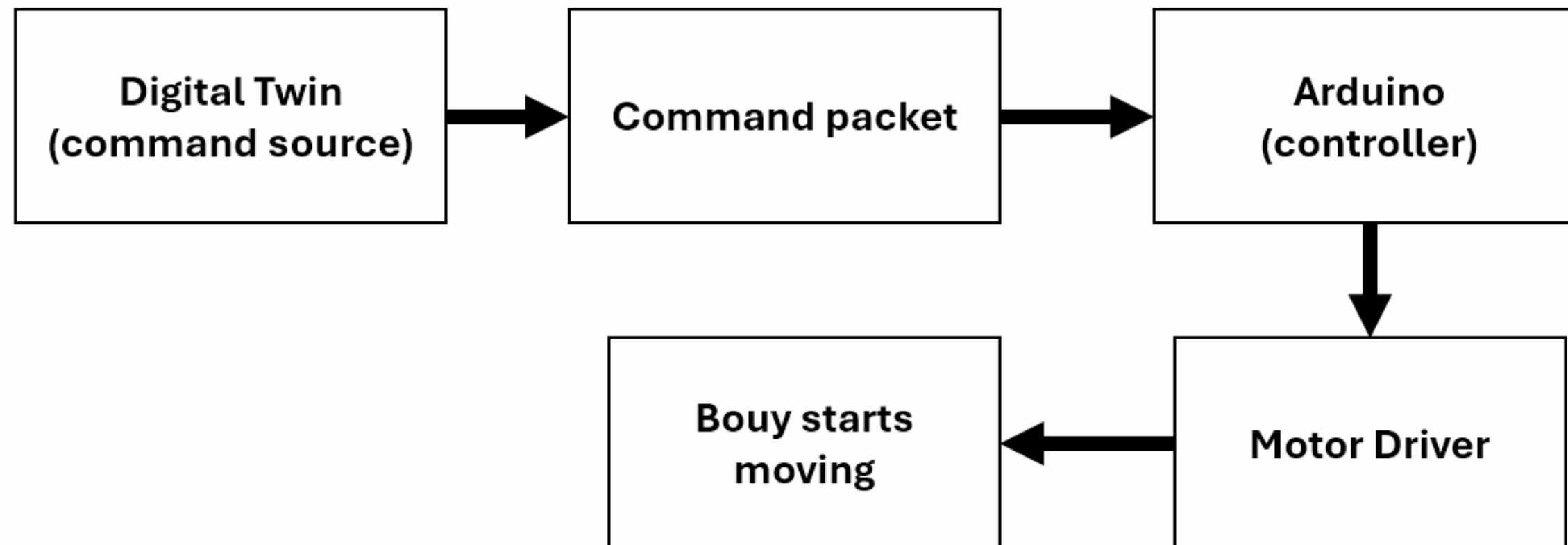


Mobility



Digital Twin to Motor Control

- simulation on Tinkercad
- commands → Arduino → motor driver → two motors
- Result: buoy can move in required direction

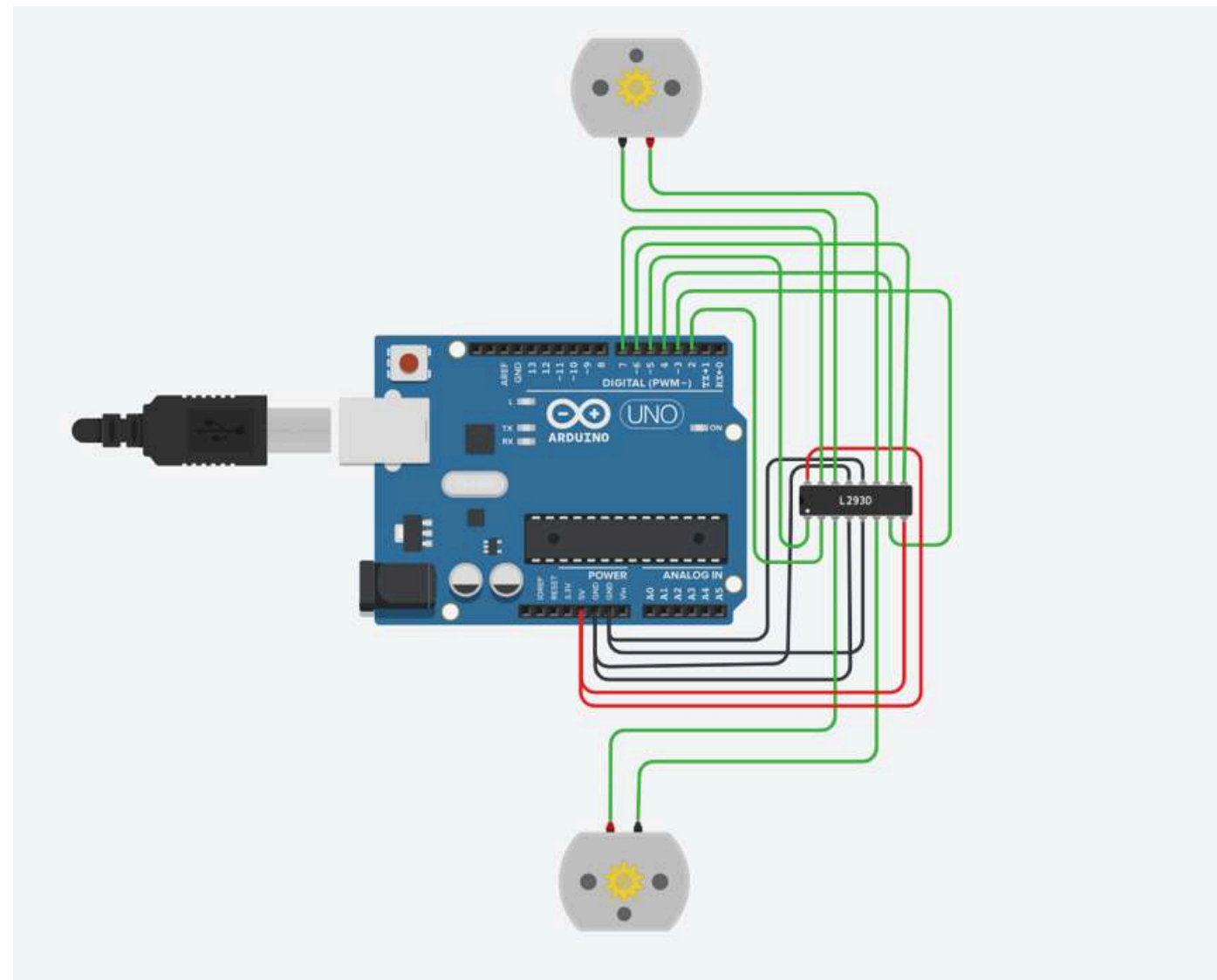


Where this fits in the CPS Buoy

- The simulation tests.
- Digital twin estimates.
- The Arduino commands.
- The motor driver powers the two motors.

Simulation setup in Tinkercad

- Arduino receives serial commands.
- Motor driver.
- Two DC Motors.



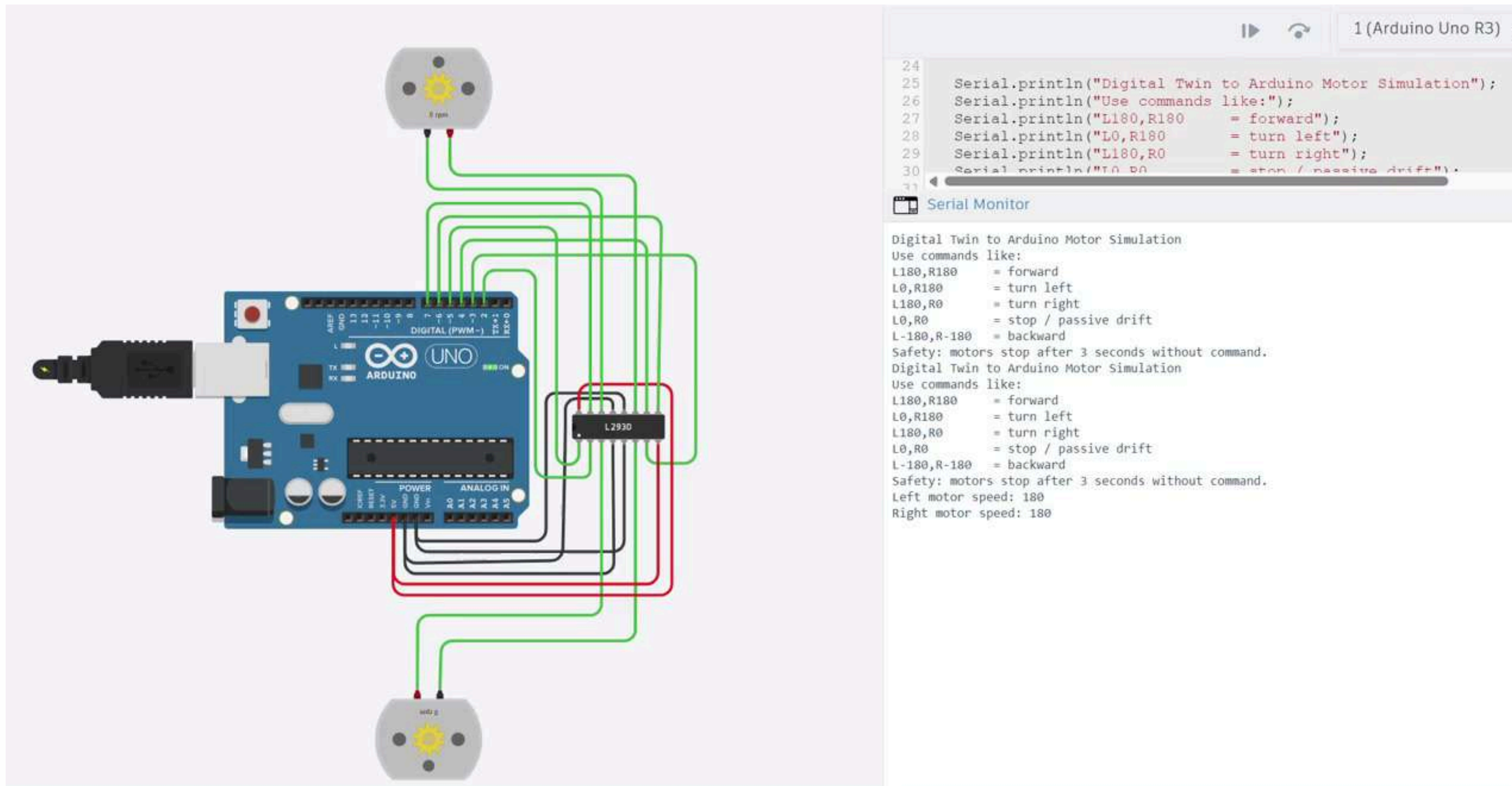
Digital-twin style commands

The command format controls each motor independently.

Command	Left motor	Right motor	Buoy action
L180,R180	Forward	Forward	Move forward
L0,R180	Stop	Forward	Turn left
L180,R0	Forward	Stop	Turn right
L0,R0	Stop	Stop	Passive drift / stop
L-180,R-180	Reverse	Reverse	Move backward
No command for 3 s	Stop	Stop	Safety timeout

Digital-twin style commands

The command format controls each motor independently.



Safety behavior added to the controller

- The motors stops when communication is lost.
- If no command for 3 seconds everything stops.
- This prevents uncontrolled movement.
- Motors are off and buoy drifts passively.
- Motors are turned ON then buoy can be guided.
- Tested result.

What was completed

Summary:

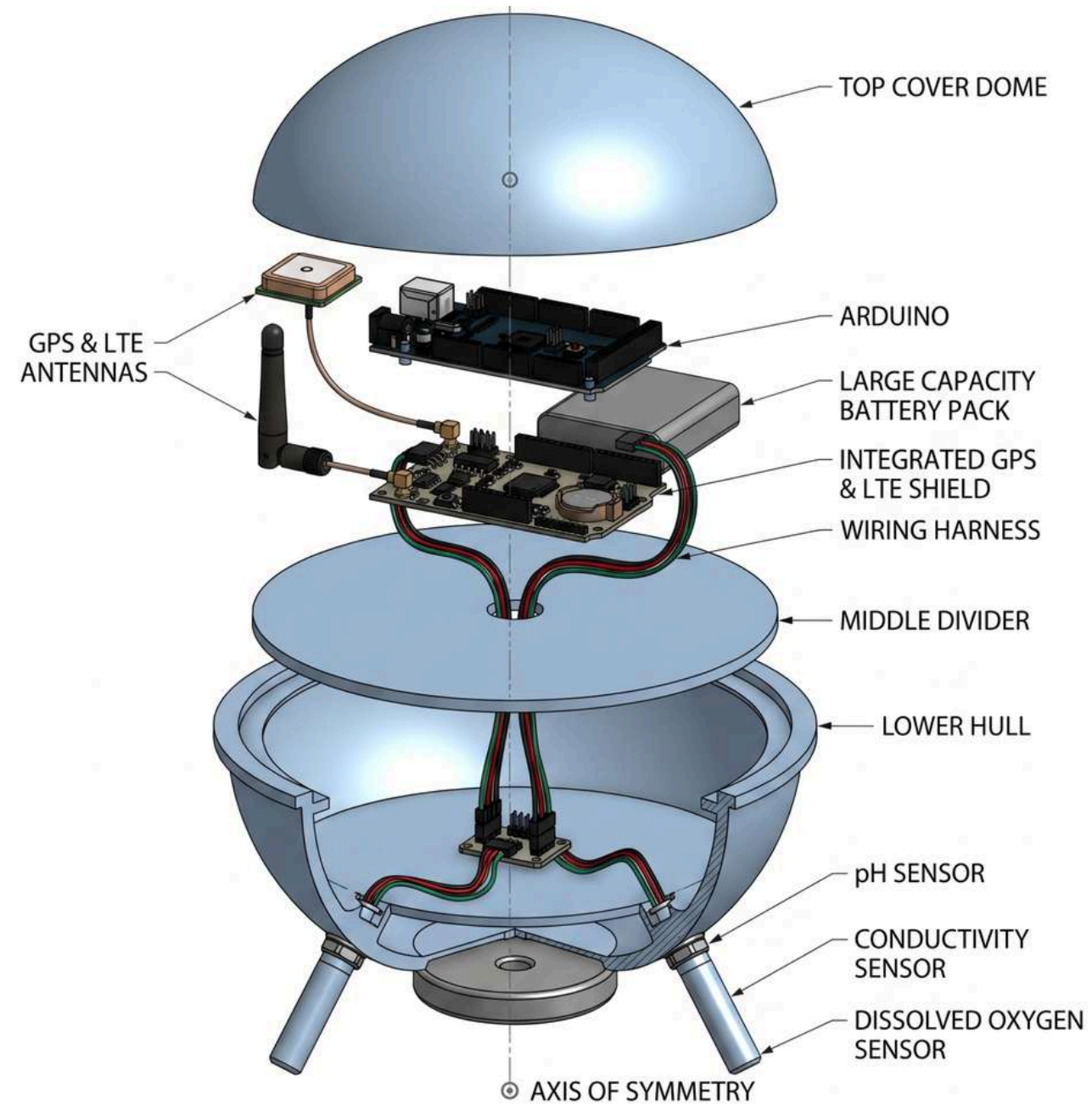
- Simulation
- Digital-twin style commands
- Validated movements
- If no command arrives.

Architecture



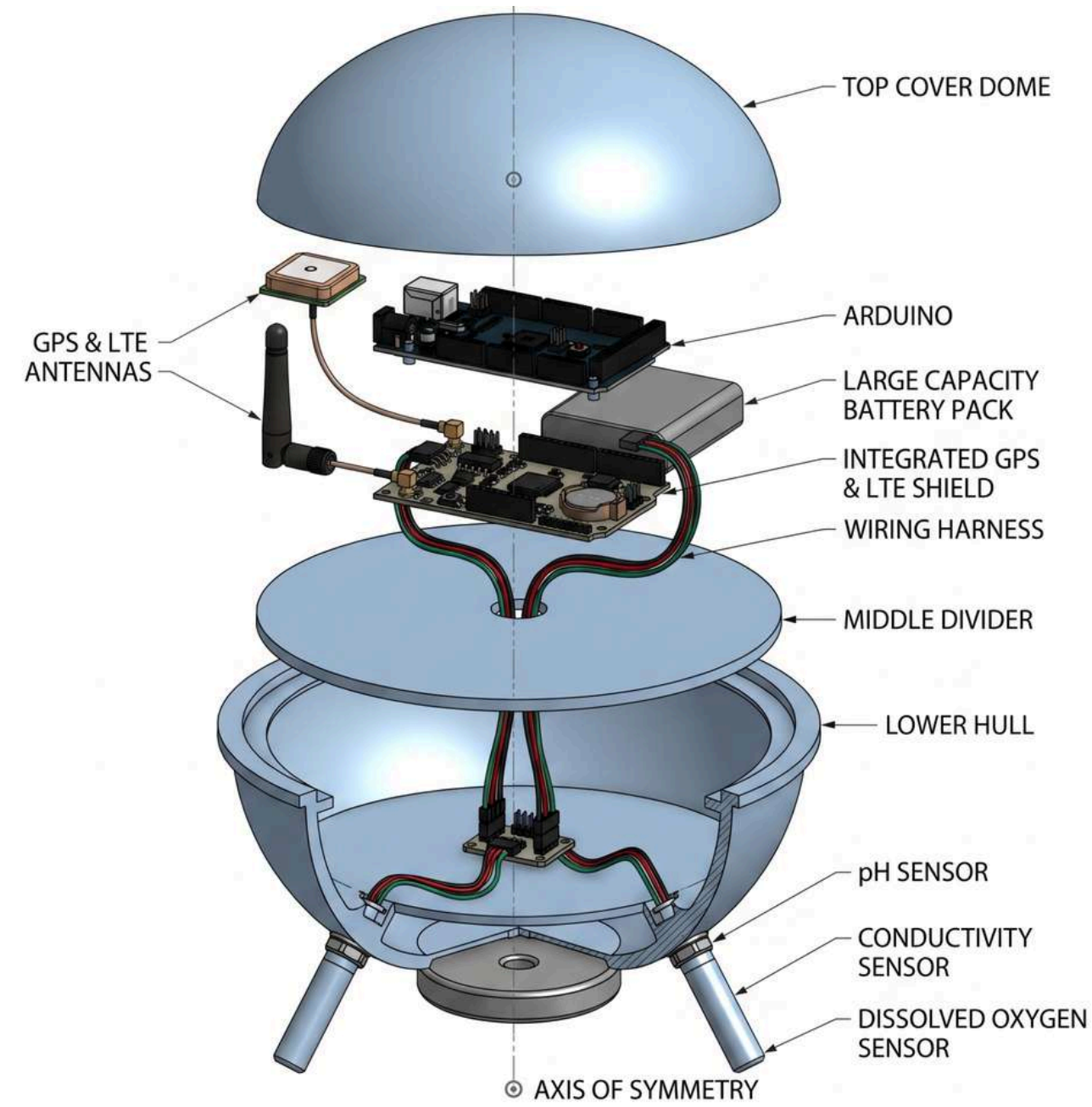
Buoy Architecture

Our Initial Plan



Buoy Architecture

Our Initial Plan

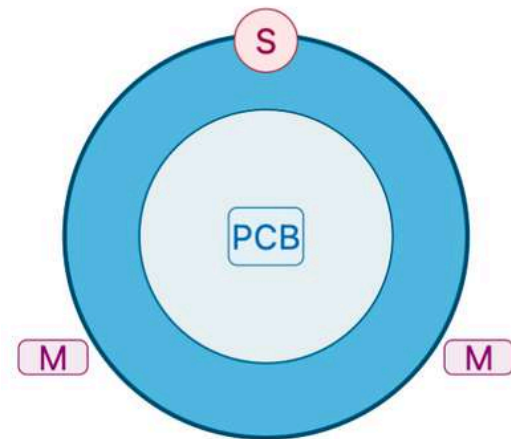


Our Final Decision

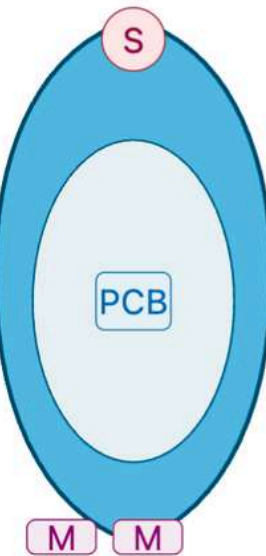


We considered 8 hull architectures

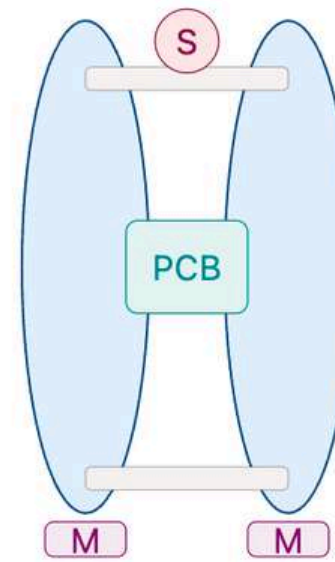
Sphere Buoy



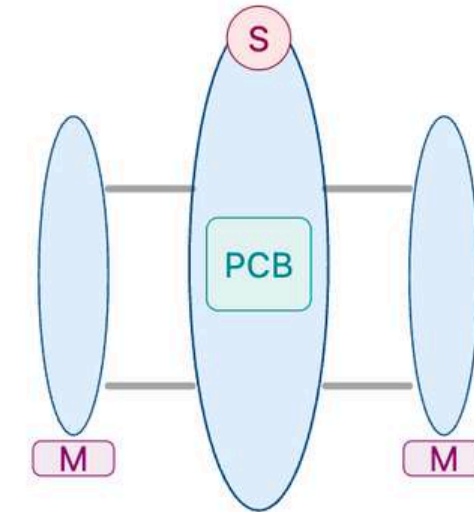
Teardrop Hull



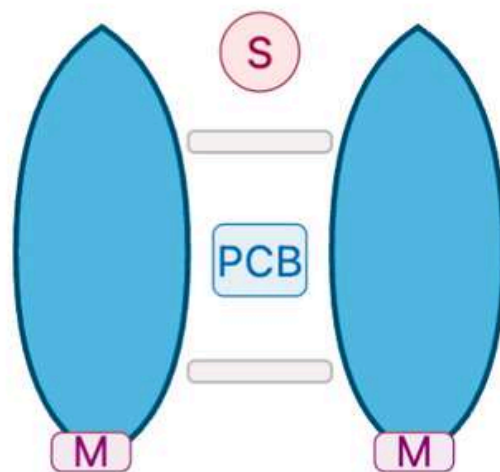
Catamaran



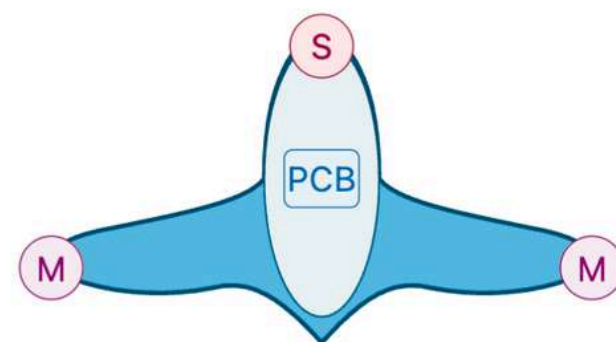
Trimaran



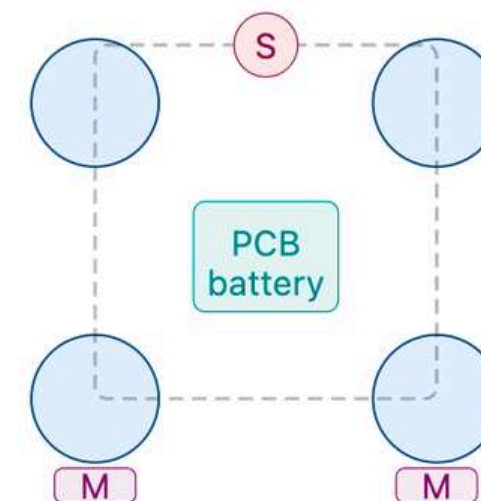
Double canoe



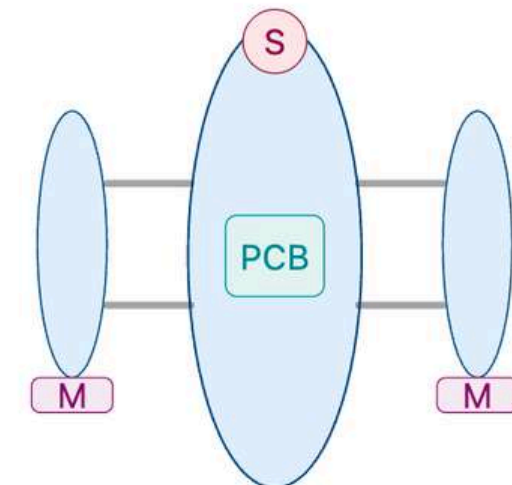
Manta Wing



Pontoon Raft



Kayak & Outriggers



We considered 8 hull architectures

Design	Roll stability	Capsize resist.	Current handling	Hydro-dynamics	Build simplicity	Sensor mounting	Electronics shelter	Best for
Disc buoy	● ● ● ● ●	● ● ● ● ●	● ● ● ○ ○	● ● ○ ○ ○	● ● ● ● ○	● ● ● ● ○	● ● ● ● ○	Slow/calm water, stationary sampling
Teardrop hull	● ● ● ○ ○	● ● ● ○ ○	● ● ● ● ●	● ● ● ● ●	● ● ● ○ ○	● ● ● ○ ○	● ● ● ● ●	River transit, long missions
Double canoe	● ● ● ● ○	● ● ● ● ○	● ● ● ● ○	● ● ● ● ○	● ● ● ● ○	● ● ● ● ●	● ● ● ● ○	Balanced structure
Manta wing	● ● ● ● ●	● ● ● ● ●	● ● ● ○ ○	● ● ● ○ ○	● ● ○ ○ ○	● ● ● ● ○	● ● ● ● ●	Showcase, low current
Catamaran	● ● ● ● ○	● ● ● ● ○	● ● ● ○ ○	● ● ● ● ○	● ● ● ● ●	● ● ● ● ○	● ● ● ● ○	Best first build, slow rivers
Trimaran	● ● ● ● ○	● ● ● ● ○	● ● ● ● ○	● ● ● ● ○	● ● ● ● ○	● ● ● ● ○	● ● ● ● ○	Moderate current, neat build
Pontoon raft	● ● ● ● ●	● ● ● ● ●	● ● ○ ○ ○	● ● ○ ○ ○	● ● ● ● ●	● ● ● ● ●	● ● ● ● ○	Calm water, heavy payload
Kayak & outriggers	● ● ● ● ○	● ● ● ● ○	● ● ● ● ●	● ● ● ● ●	● ● ● ○ ○	● ● ● ● ○	● ● ● ● ●	Strong current, long range

To build or to buy?

Build - 3D-print custom hull

1. ~ Three weeks print and reprint cycle
2. Waterproofness is not guaranteed
3. Single point of failure if it leaks
4. Not enough specialization

Buy - RC bait boat

1. Catamaran hull already manufactured-optimized
2. Payload bay built in for our PCB and sensors
3. Waterproof from the factory

Total Cost: € 360



What we purchased and why



Catamaran - like Shape

Matches our trade-off winner from the previous slide

Payload bay

Houses the components without modification.

Dual motors

Dual water-cooled
550-size DC motors
FR4 satisfied

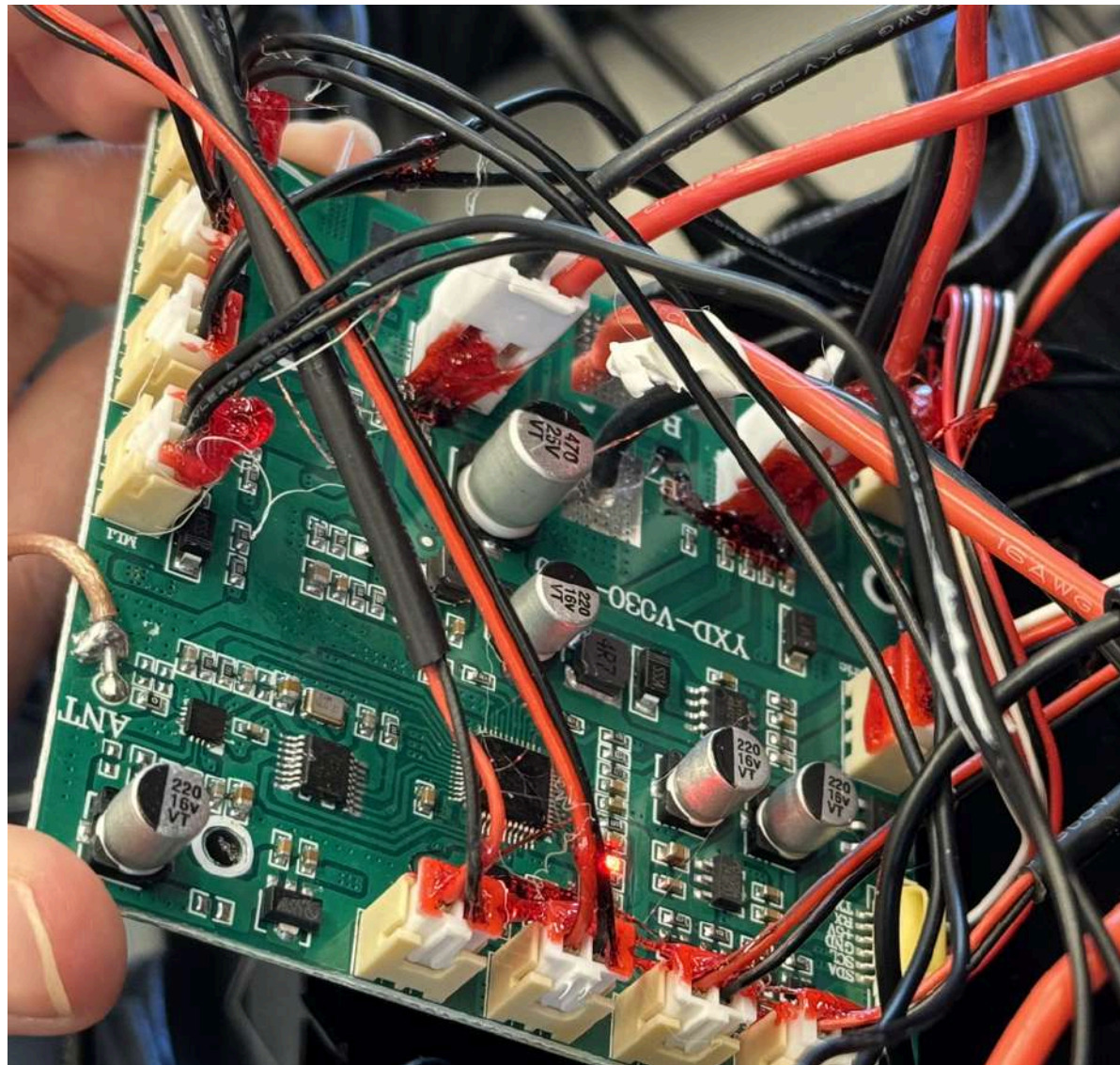
RC remote

NFR2 (human emergency stop) already provided.

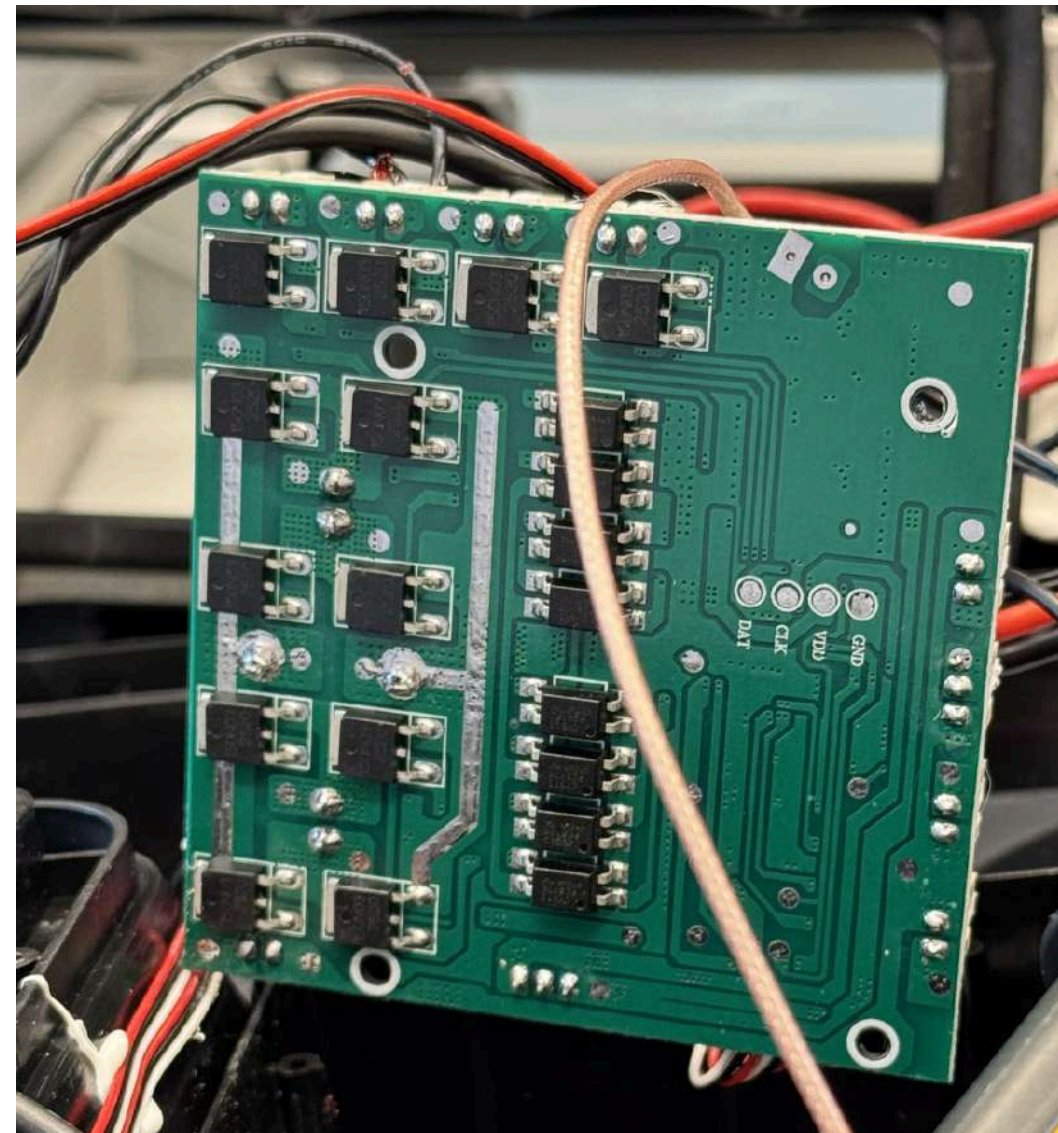
4 of 16 requirements satisfied

The problems that came with the purchase

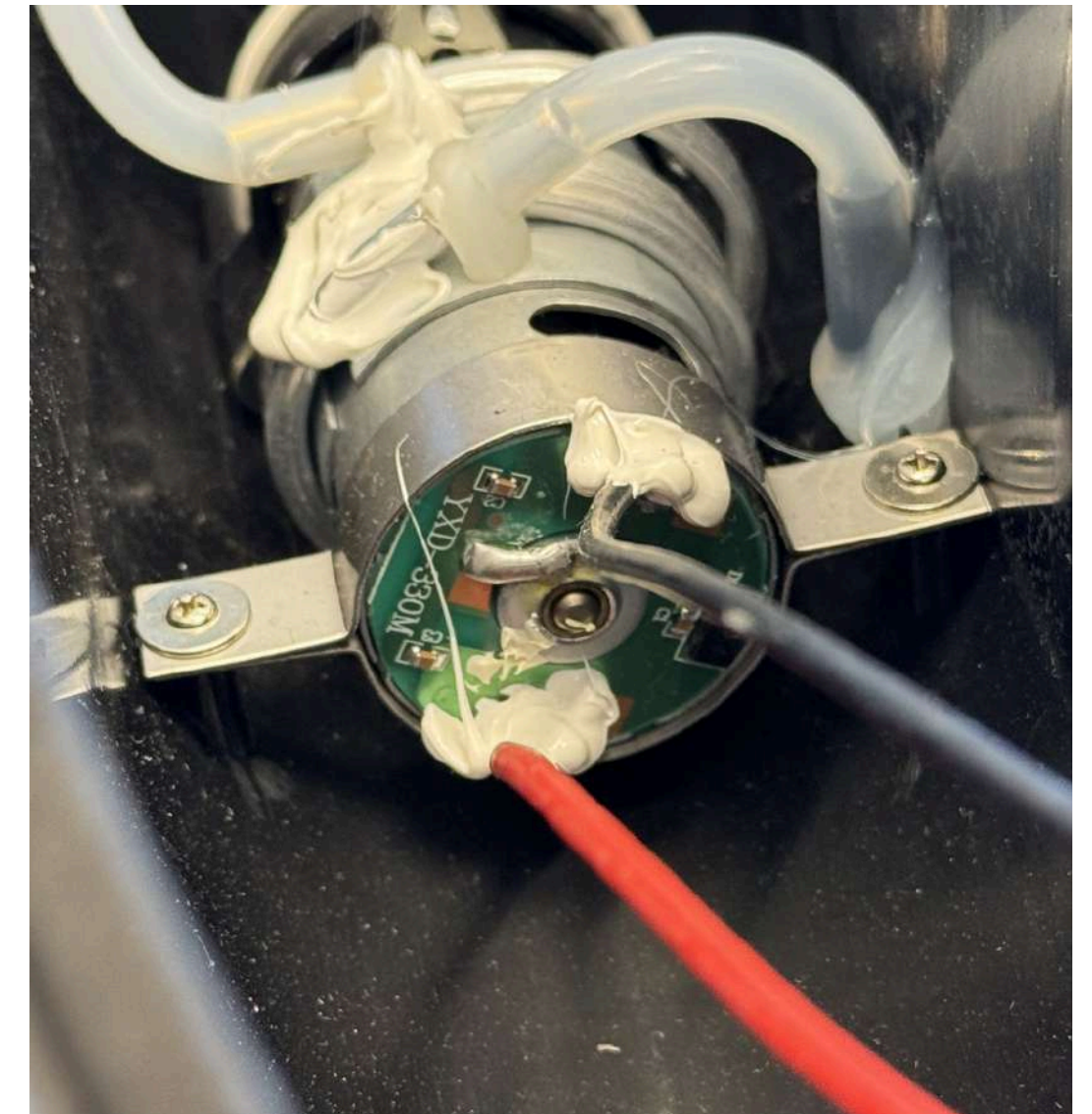
Limited space inside



Custom vendor PCB



Everything glued



A boundary instead of a shortcut



No public documentation

PCB is Custom-made by the manufacturer.
No schematic, no pinout, no datasheet.



Reverse-engineered partial

Probed pins with a multimeter
mapped power, ground, and a few signal lines
but not enough for interfacing.



Firmware incompatible

Arduino library expected logic levels and protocols
that the boat MCU did not provide.

*Reverse-engineering a
vendor PCB for marginal
benefit is a rabbit hole.*

How we solved it

Not integrate in a single circuit

	Arduino Mega	ESP32
Physical size	Large (10 × 5 cm)	Small (5 × 3 cm)
GPS/ LTE	External shield	Native
Power consumption	200 mA active	80 mA active

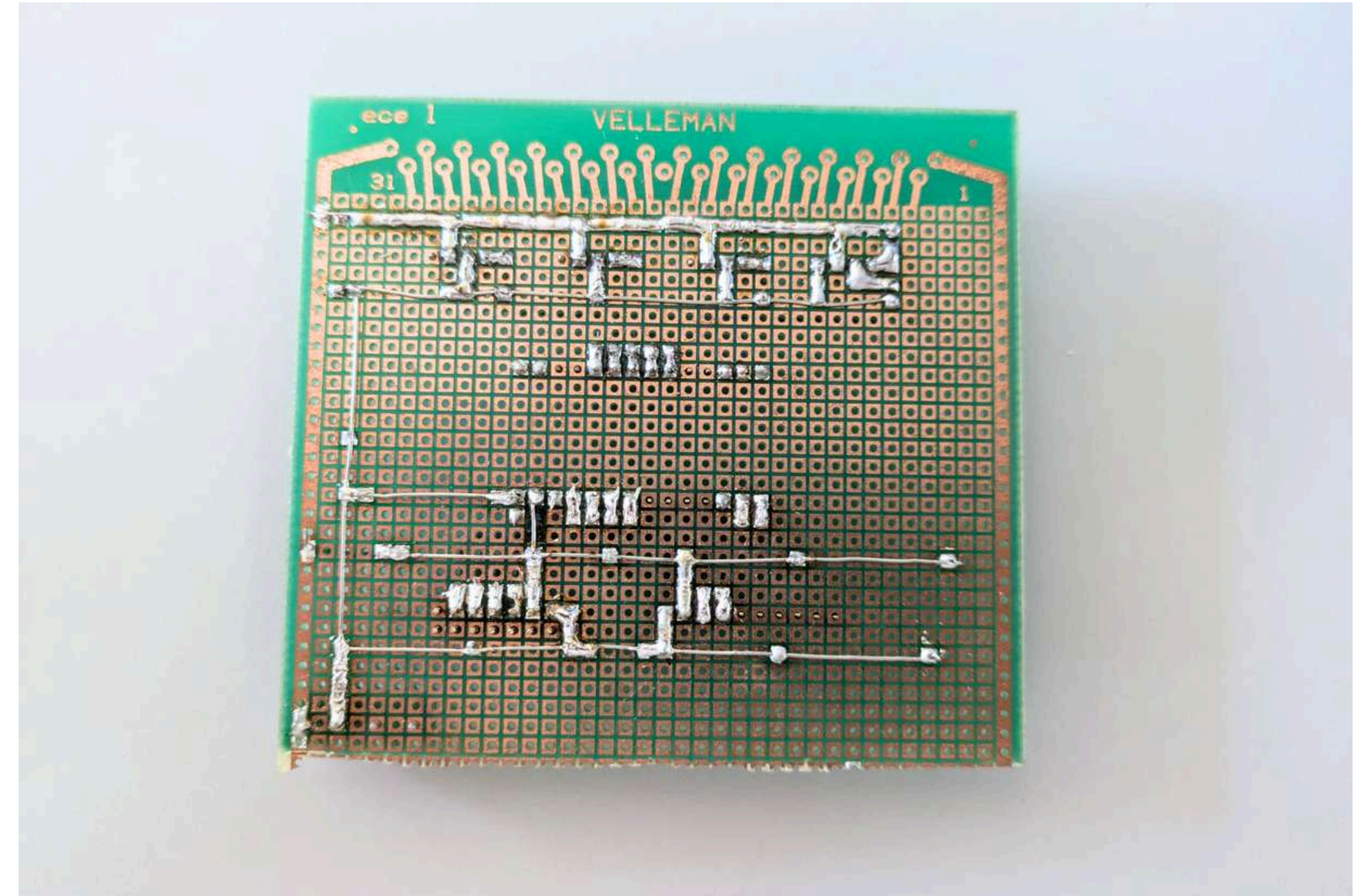
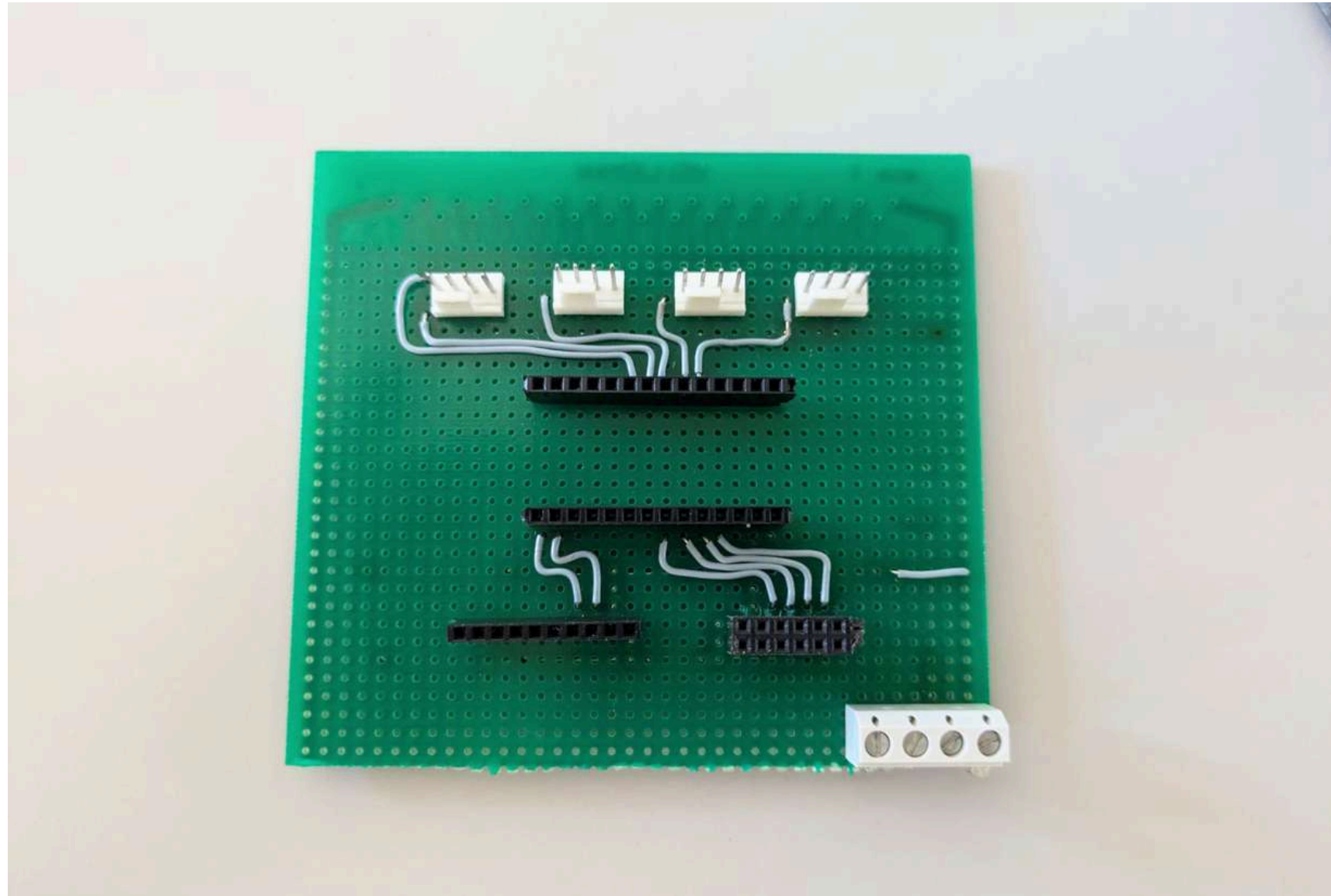
- Place our MCU and boards on payload area
- Keep the remote control
- Send commands to the motor via ESP32 MCU

Implementation



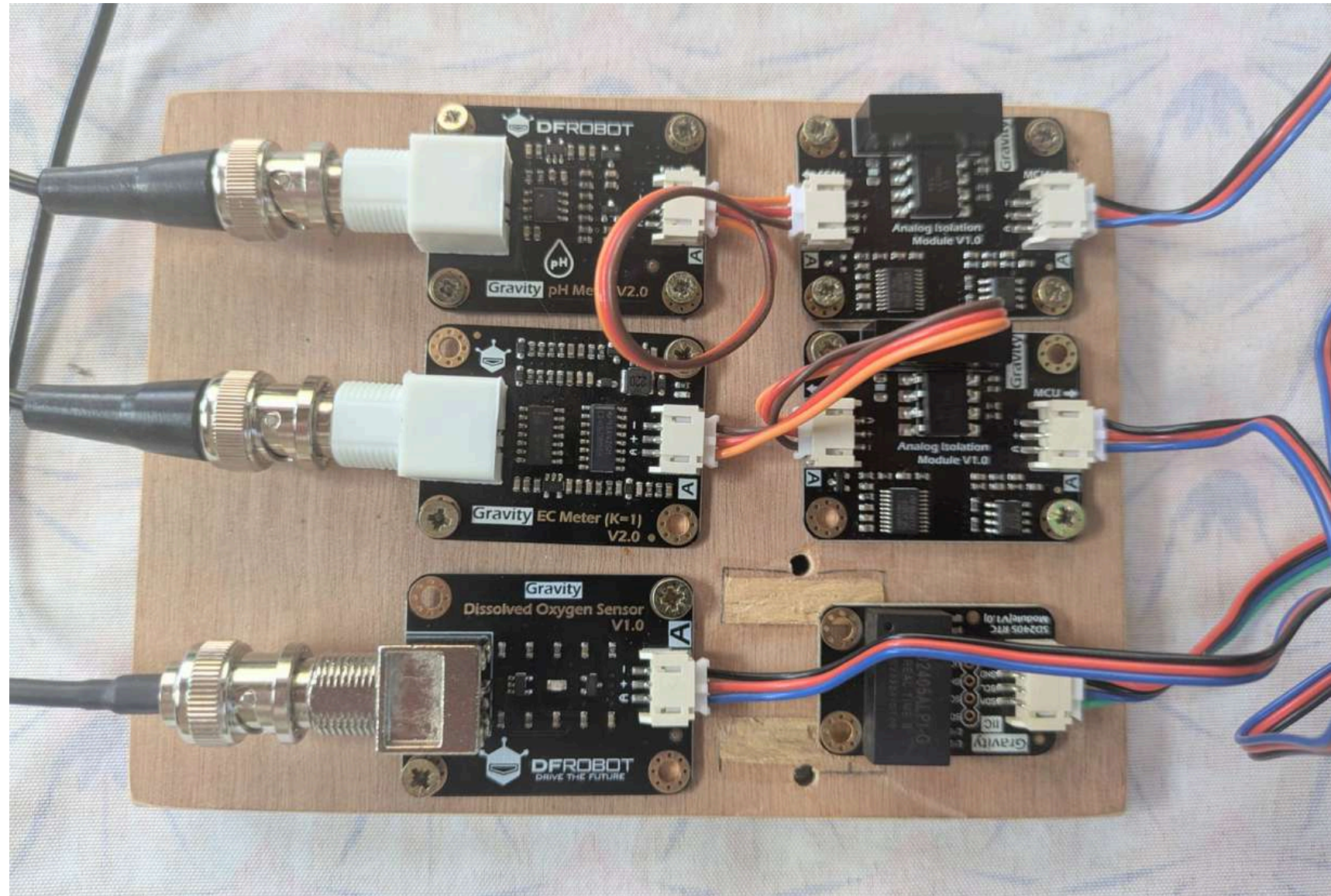
Time to clear the mess

Made our own PCB

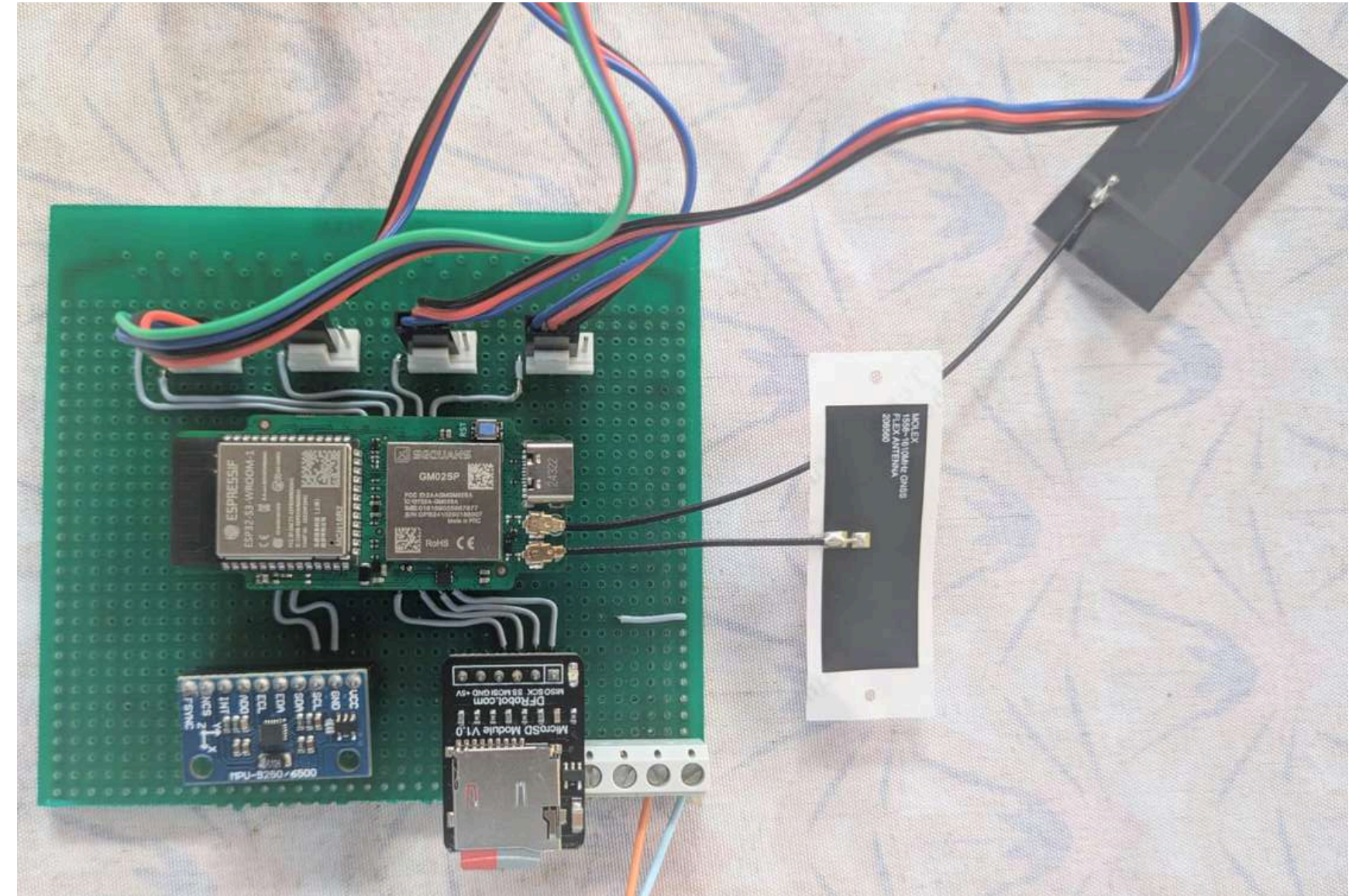


Time to clear the mess

Sorted the sensor boards



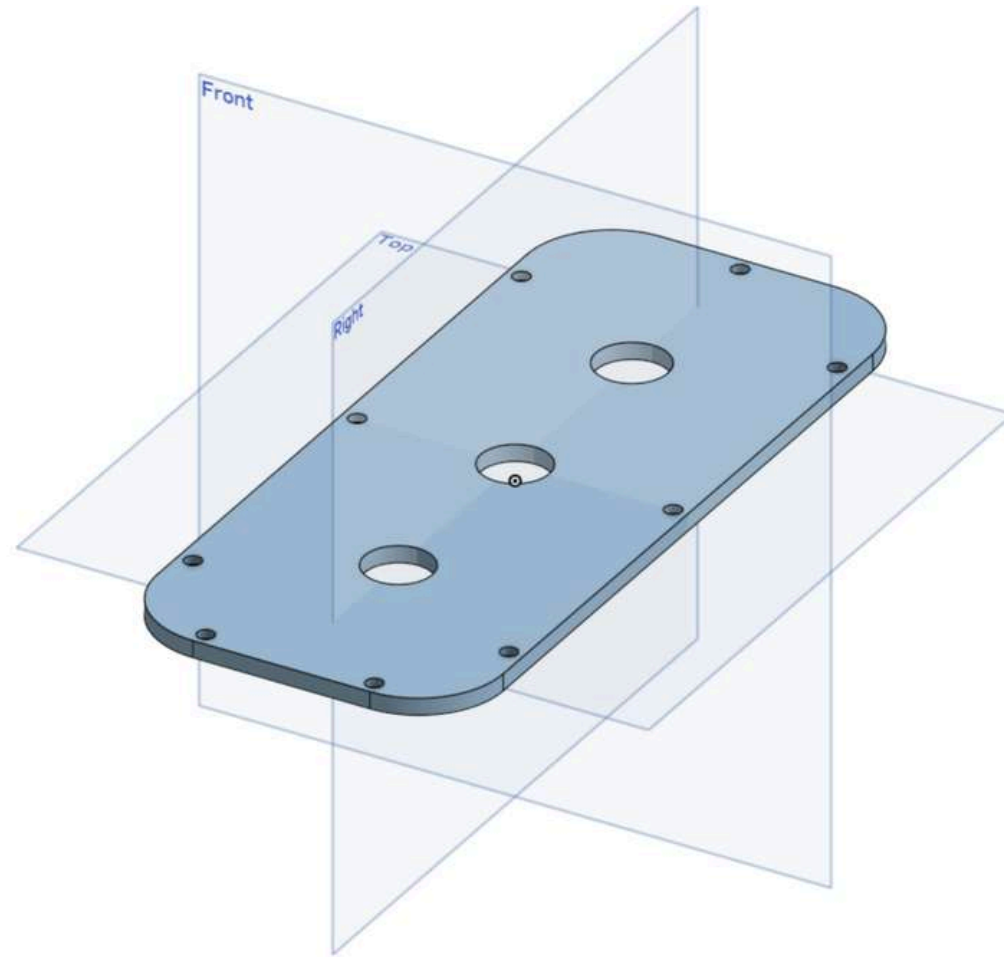
Connected the components



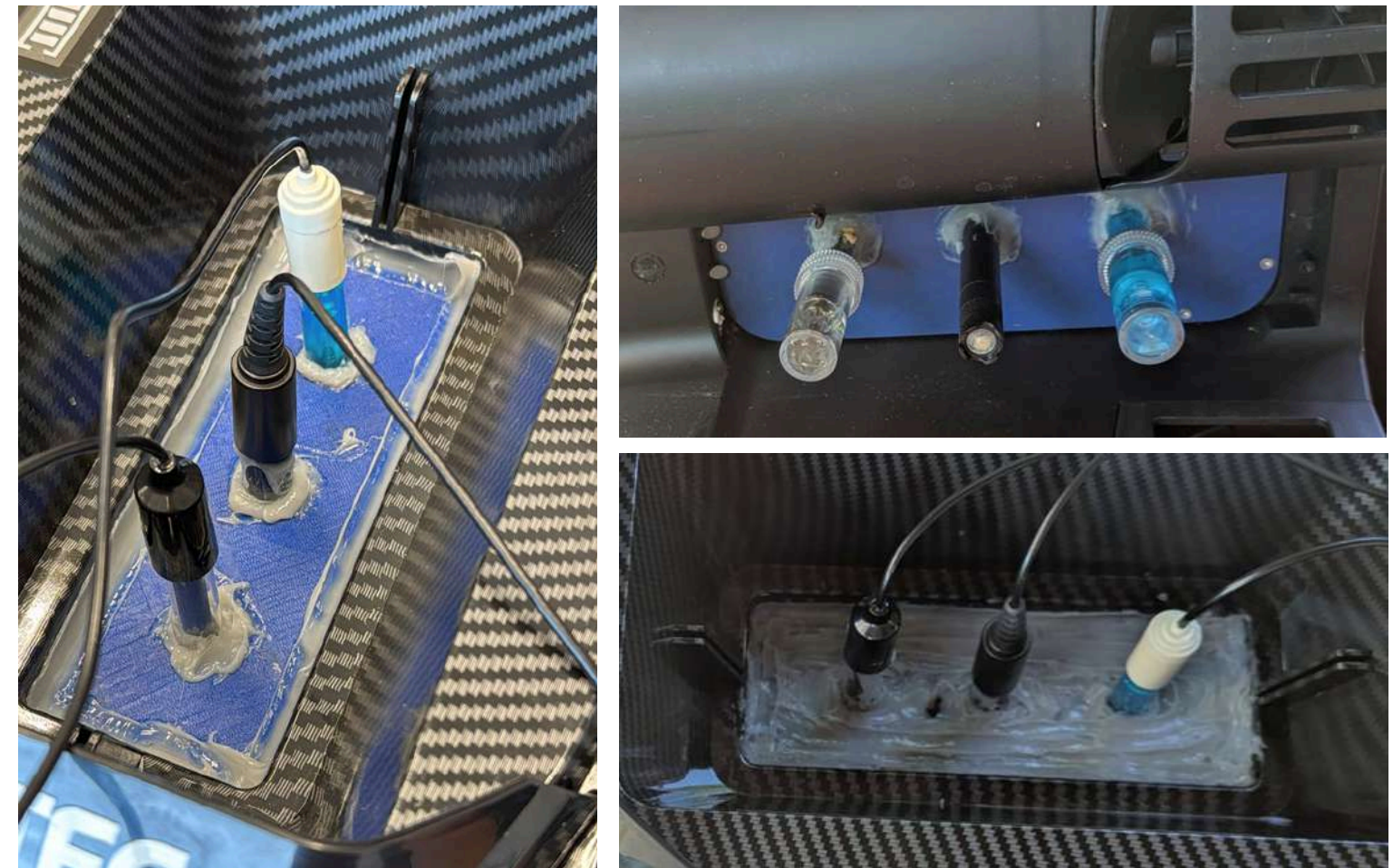
Keeping the electronics dry

Custom-measured, 3D-printed in-house, zero modification to the boat.

Designed



Installed



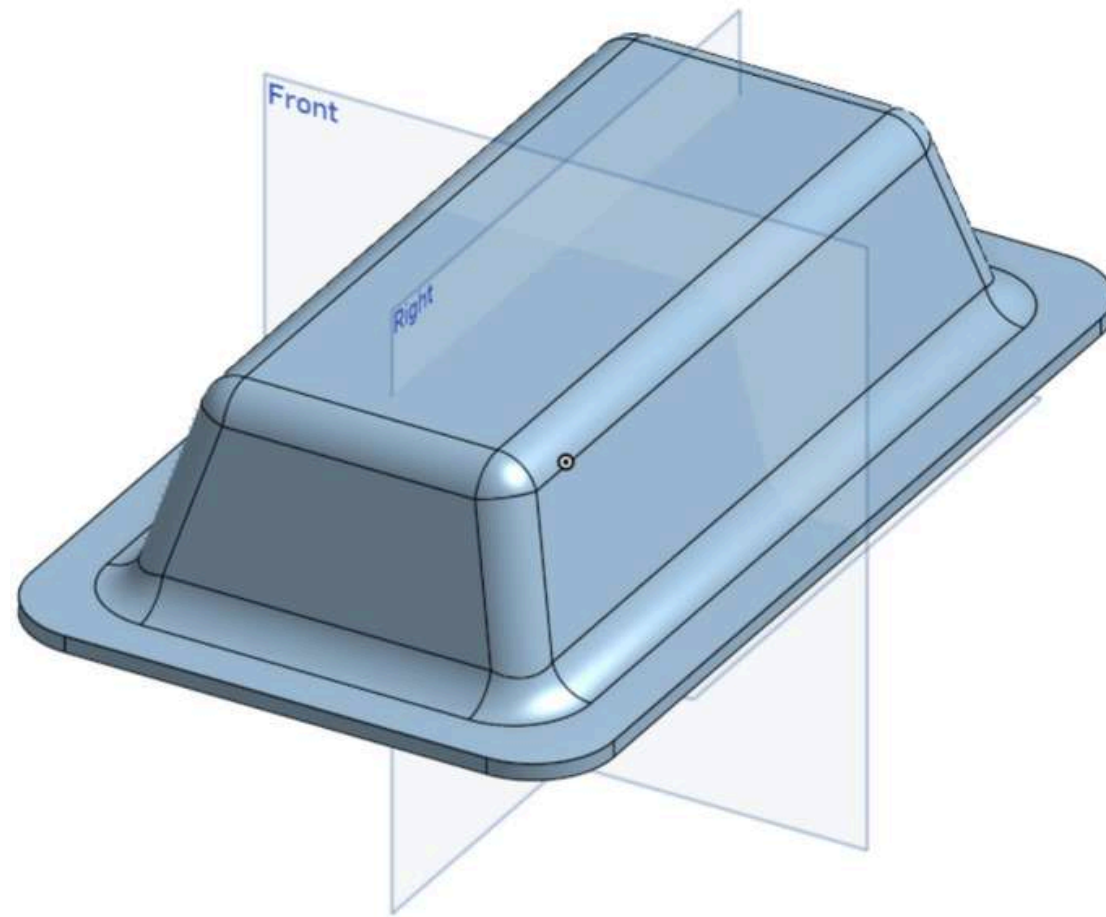
Sensor probes extend through the 3D printed part.

FR2 (underwater sensor protection) Satisfied

Keeping the electronics dry

Custom-measured, 3D-printed in-house, zero modification to the boat.

Designed



Installed



The dome-like hull protects the electronics
NFR4 (Waterproof top) Satisfied

Added inertial measurement unit (IMU) sensor

Why we need it

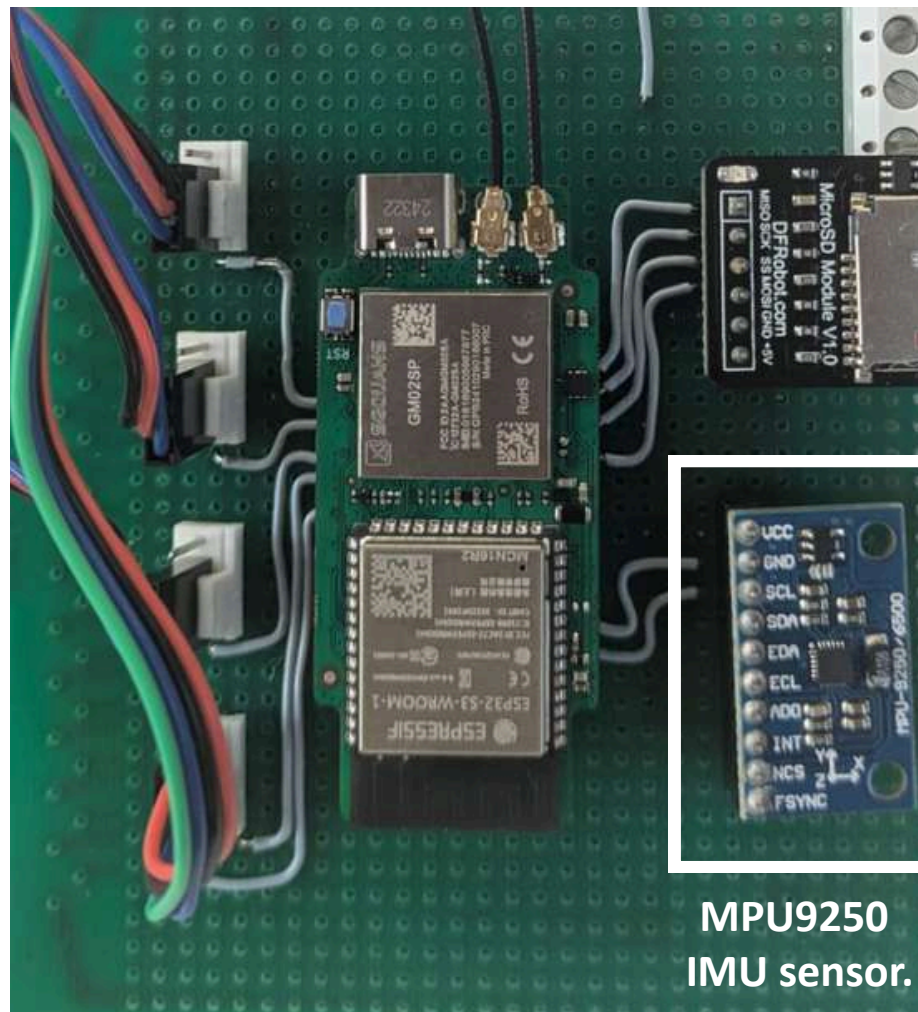
Reports 3-axis heading, acceleration, and angular velocity

The digital twin can use this data for pose estimation

How it will work (future)

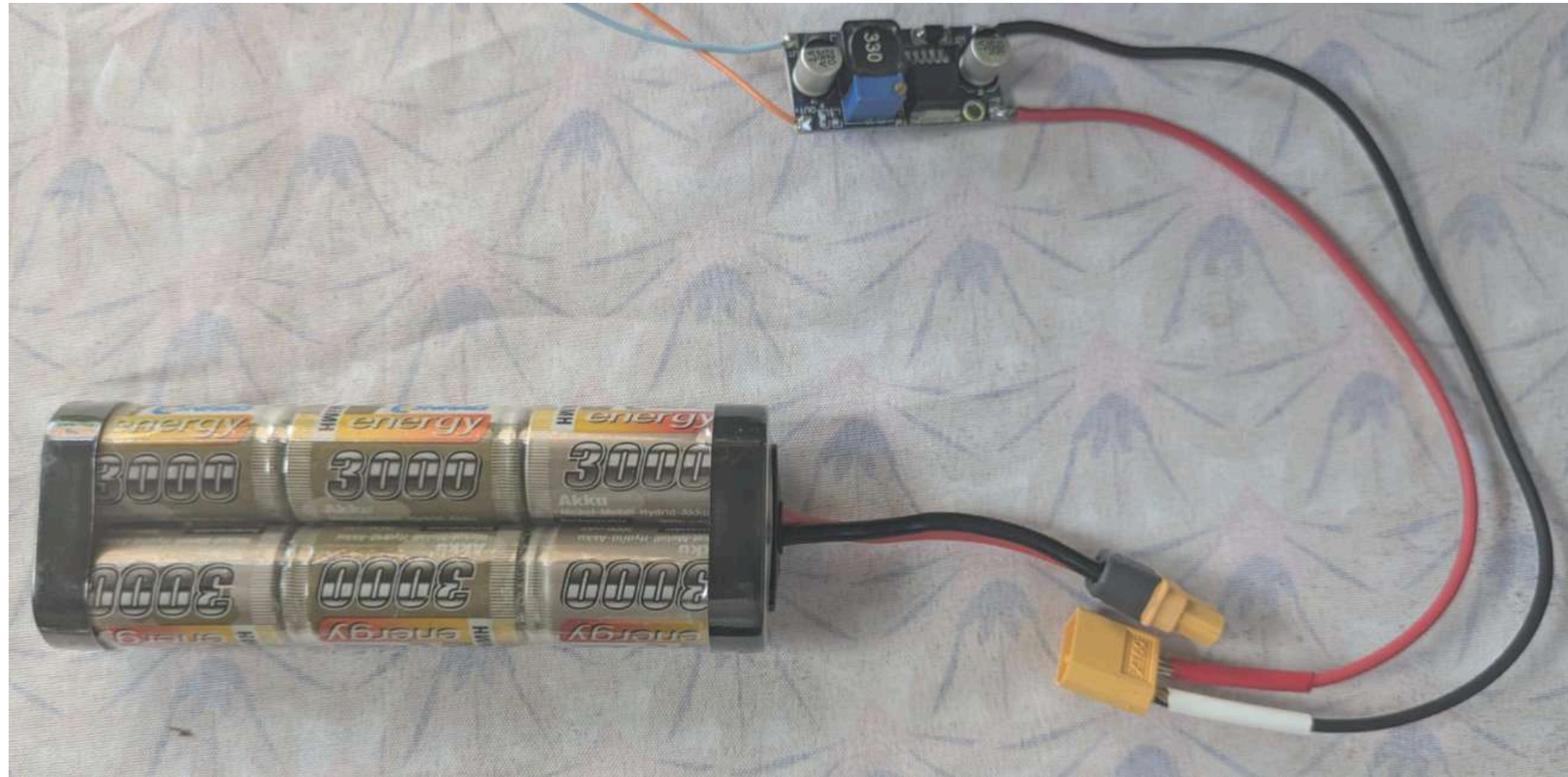
Closed-loop control:

- Digital twin computes the next best heading
- sends RC-equivalent command
- boat steers.



FR6 evolved - We add IMU sensor to our sensor list

Power Supply



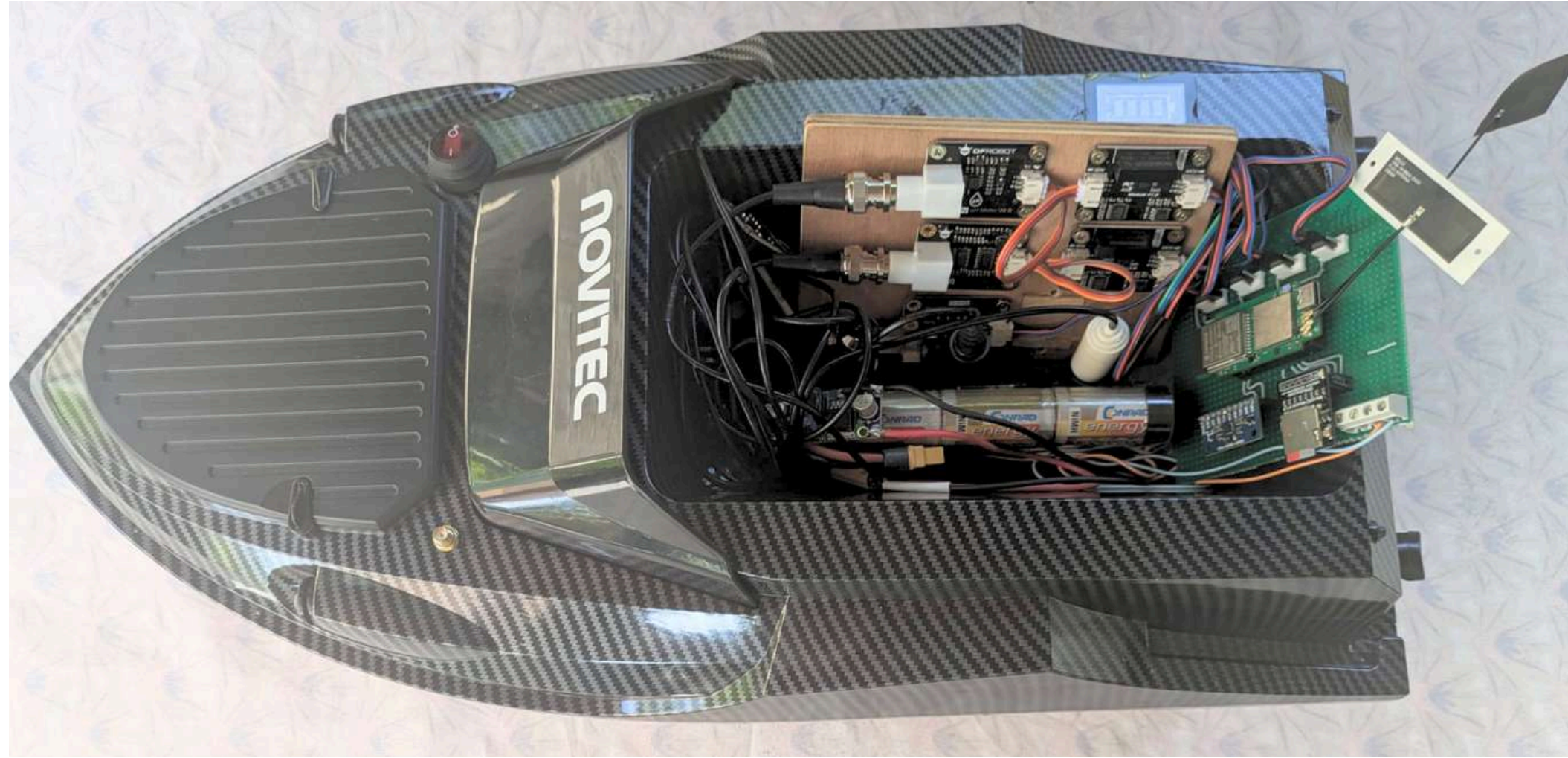
Battery Pack + DC-DC Converter

Sufficient for a 4h operation

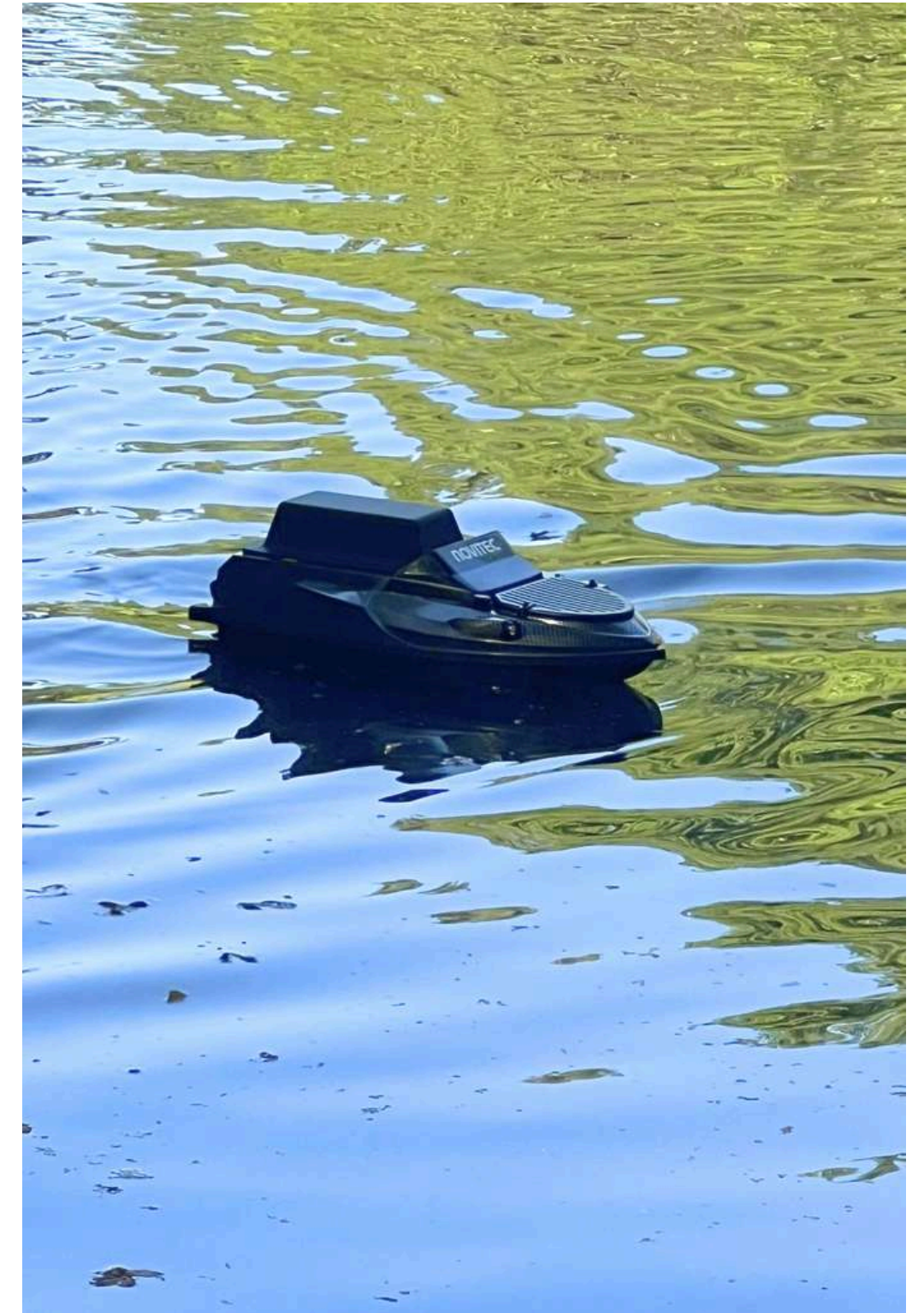
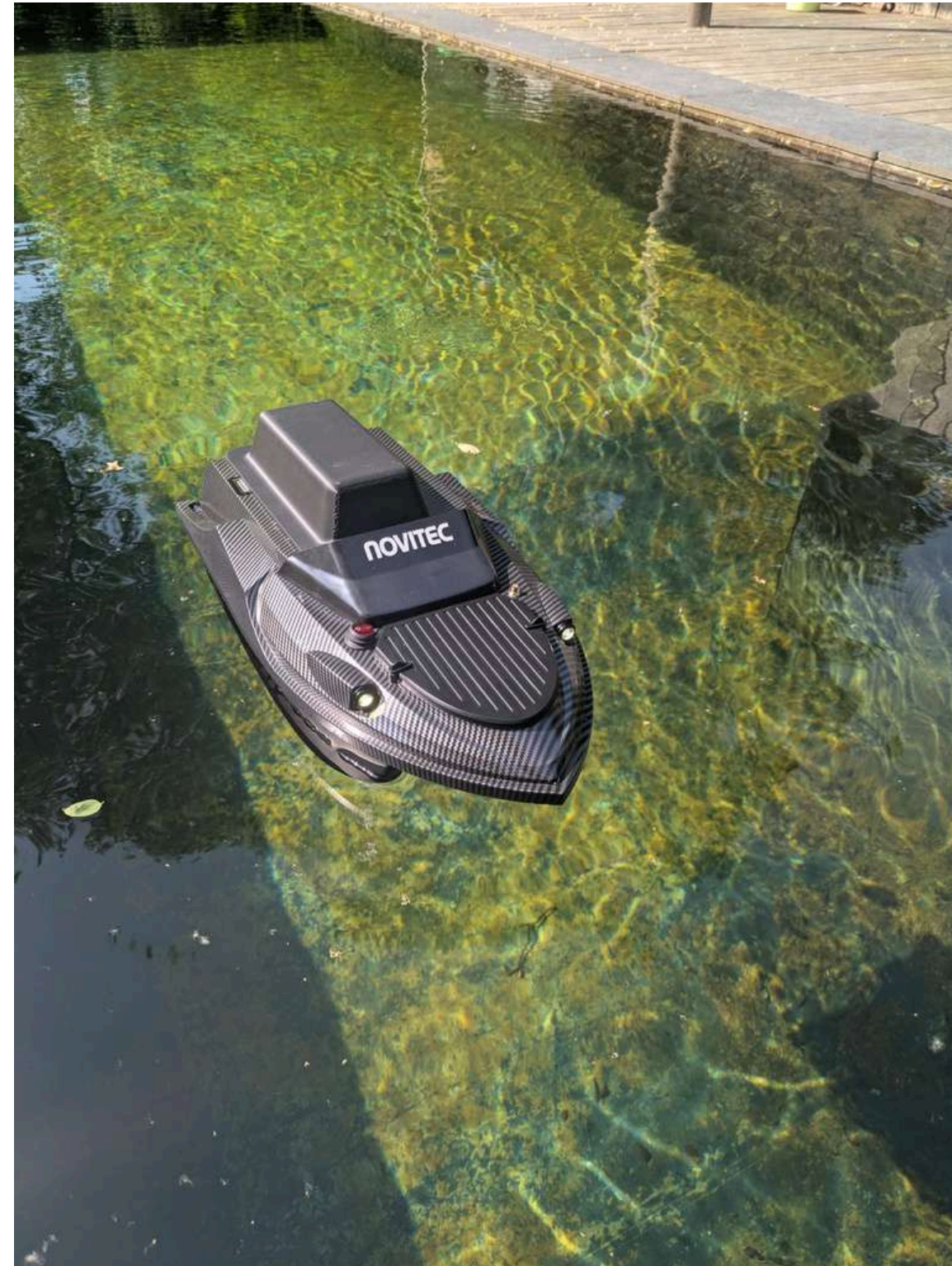
Boat motors & control work on separate battery

- 6 × NiMH cells
- 7.2V nominal 3000 mAh
- Total Energy: 21.6 Wh
- ~ 27 hours of runtime
- NFR3 is satisfied.

Boat Assembly



Boat in Action



Boat in Action



Digital Twin





Digital Twin

A mirror of the physical buoy extended with river physics and an source estimator.

Data input

(pH, EC, DO, GPS, IMU)
Interfaces with
thingsboard
or simulated values

Interpretation

River hydrodynamics model
and anomaly detection

Localize

Runs Bayesian inference
to estimate pollution source

Three Modes

SIM (synthesized sensors)

REAL (live buoy telemetry)

HIL (controller test)



Architecture



River Model

Curvilinear* grid from user-drawn centerline or OSM data.

Flow velocity per cell from open-channel hydraulics.



Detection logic

River-specific thresholds.

Alarm history logged with position, severity, and concentration.



Source estimator

Bayesian inference over all candidate source cells on the grid.

BUOY

DIGITAL TWIN - MEUSE

LIVE SENSOR DATA

Temperature

32.60°C °C

Latitude

51.177864 °

Longitude

4.414857 °

pH

4.57

EC

15.10 μS/cm

DO

4.76 mg/L

Last update

03:42:37

BUOY POSITION (LOCAL)

Stream X

820.42 m

Cross Y

-94.22 m

Sim time

693 s

MODE

Simulated flow

Real GPS (live)

RIVER PLACEMENT

Draw RIVER centerline

Draw river WIDTH

Place SOURCE on map

Place BUOY START on map

Reset river (use default)

river: 498 pts | width: 80.0 m

Profile name

SAVE

Load saved river...

LOAD SELECTED





Contamination Source Simulation

Governing equation solved on the curvilinear* grid with anisotropic dispersion :

$$\frac{\partial C}{\partial t} + u \frac{\partial C}{\partial x} + v \frac{\partial C}{\partial y} = \frac{\partial}{\partial x} \left(D_L \frac{\partial C}{\partial x} \right) + \frac{\partial}{\partial y} \left(D_T \frac{\partial C}{\partial y} \right) + S$$

Longitudinal dispersion

$$D_L = 12 \text{ m}^2/\text{s}$$

Transverse dispersion

$$D_T = 2.5 \text{ m}^2/\text{s}$$

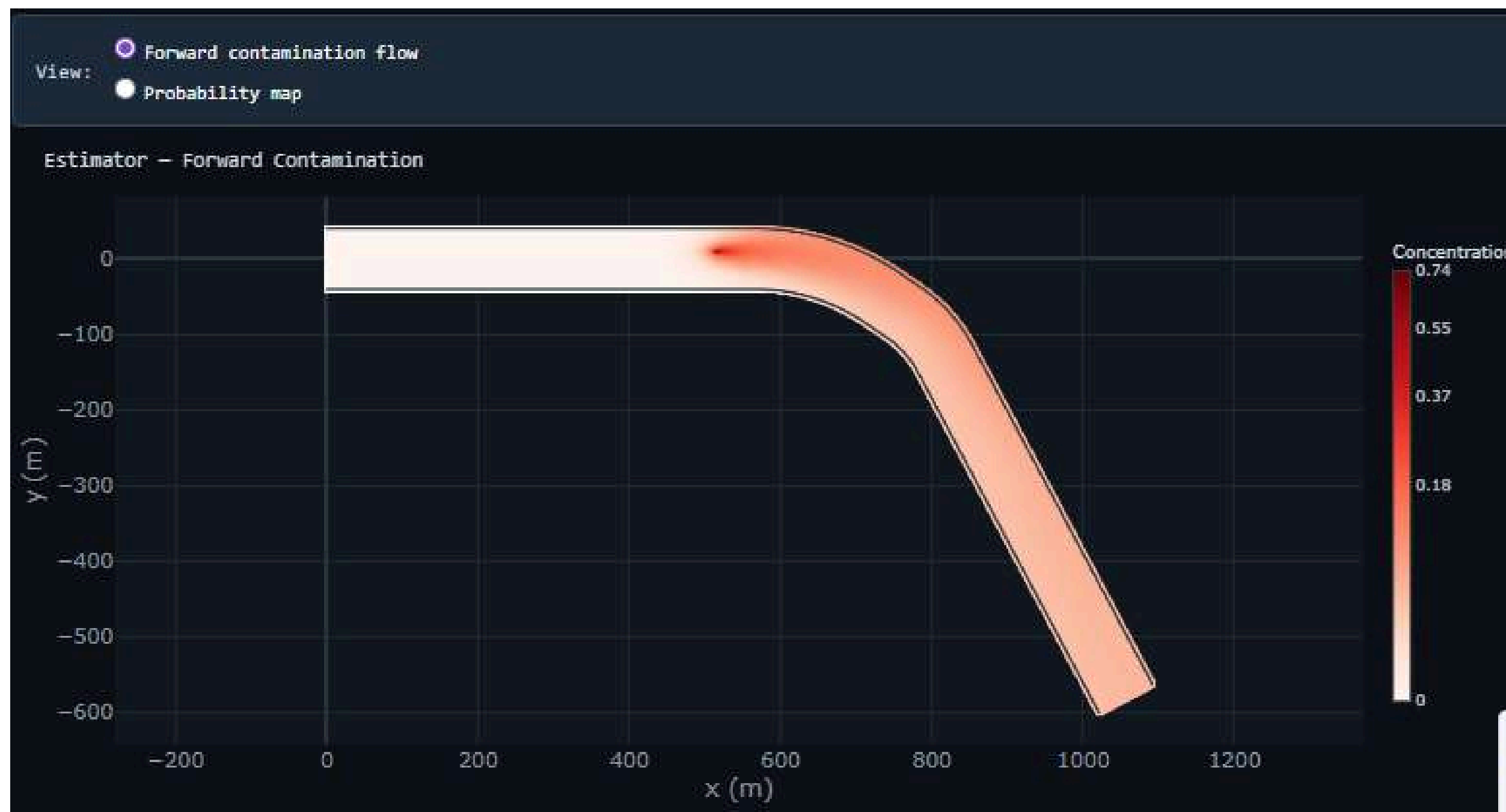
Solver

Implicit Finite Volume
Sparse LU decomposition

Pollution is a long thin streak hugging the release bank, not a blob



Contamination Source Simulation





Source Estimation

Find where pollution came from?

1 Forward shape function

Analytic Gaussian plume predicts expected concentration at each observation point, for every candidate source cell.

2 Eliminate source strength Q analytically

$$Q^*(s) = \sum g_k(s) \cdot c_{\text{obs},k} / \sum g_k(s)^2$$

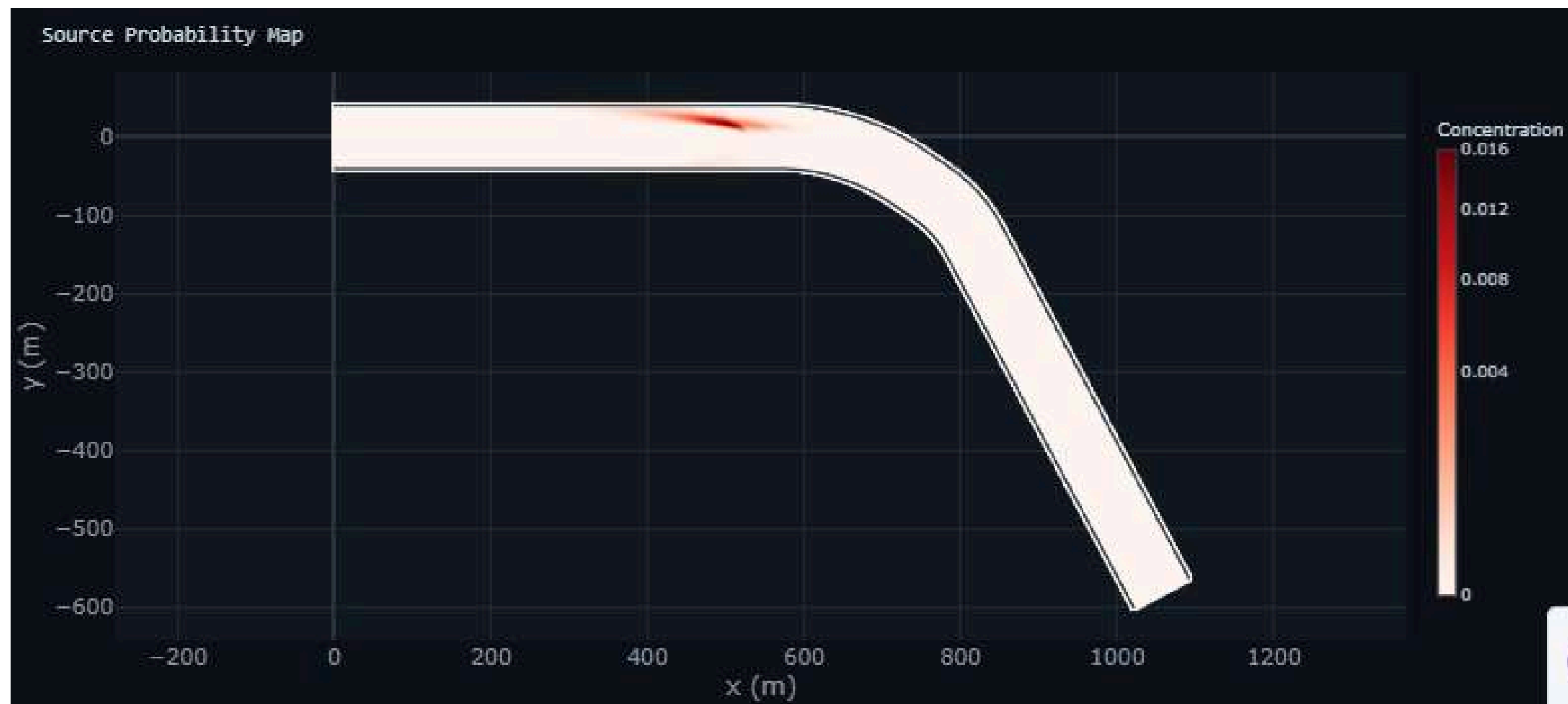
g = expected concentration per unit source

3 Posterior probability map

Profile log-likelihood converted to a probability map over all river cells via softmax.
Non-detections penalize candidates whose pollution would have been seen.



Source Estimation



Live Demo at the end



Return to Requirements



Return to Requirements - Functional

ID	Promise	Status	Evidence
FR1	Real-time monitoring, 10 s stability	Met	Lab calibration curves
FR2	Underwater sensor protection	Met	3D-printed mount
FR3	Passive drifting during sampling	Met	Boat motors idle, drift only
FR4	Motorized recovery	Met	Native bait-boat motors
FR5	GPS < 5 m, timestamped	Met	SoM field test
FR6	Digital twin integration	Refined	Extended with IMU heading
FR7	Source estimation	Met	Backtracking in DT
FR8	Automatic alerts	Met	Meuse-zoned thresholds
FR9	Offline data storage (SD)	Met	SD slot addition
FR10	Consistent data format	Met	Thingsboard JSON payload

Return to Requirements - NonFunctional

ID	Promise	Status	Evidence
NFR1	Failure detection within 30 s	Met	Thingsboard and digital twin
NFR2	Emergency stop within 1 s	Met	Boat RC override
NFR3	4 hour autonomy	Met	NiMH cells and separate battery for motors
NFR4	Waterproof, upright	Met	Insulating hull and 3D mount
NFR5	Environmental & Weather conditions	Met	Waterproof and insulated boat
NFR6	Cost $\leq$ €1000	Met	Final Cost $\approx$ €870

Future work

Future Work

Closed-loop control

Digital twin can send boat heading commands.

The IMU already provides the feedback signal we would need.

Estimated effort: 3 weeks.

Multi-buoy fleet

Multiple buoys cooperate to localise faster.

Thingsboard already supports multi-device dashboards.

Real - Pollution Testing

Requires special permits
Digital Twin three modes test

Other:

Fail-safe mode
Collision detection and avoidance
Depth - aware control

Demo

